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Security camera device with passive infrared (PIR) sensors

Abstract

An audio/video (A/V) device includes a first housing and a second housing that rotationally couples and uncouples to the first housing. The first housing includes a frame having a first cavity, a second cavity, and a wall disposed between the first cavity and the second cavity. The first housing further includes a mirror that extends between the first cavity and the second cavity, a first passive infrared (PIR) sensor at least partially disposed within the first cavity, and a second PIR sensor at least partially disposed within the second cavity. The second housing includes a loudspeaker.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
11393309	12/2021	Wang	N/A	G08B 13/19
2003/0219134	12/2002	Su et al.	N/A	N/A
2010/0128123	12/2009	DiPoala	N/A	N/A
2016/0364009	12/2015	Lemberger	N/A	N/A
2017/0280109	12/2016	Scalisi	N/A	H04N 23/667
2018/0322751	12/2017	Sacre et al.	N/A	N/A
2018/0357871	12/2017	Siminoff	N/A	G08B 13/189
2020/0173617	12/2019	Bancken et al.	N/A	N/A
2022/0271422	12/2021	Grillo	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
19821302	12/1998	DE	G01P 13/00

OTHER PUBLICATIONS

Office Action for U.S. Appl. No. 17/953,780, mailed on Jun. 14, 2024, Gilbert, “Security Camera Device”, 16 pages. cited by applicant

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Background/Summary

BACKGROUND

(1) Security is a concern for many homeowners, businesses, and the like. In some examples, those seeking to protect or monitor their homes or place of business may wish to have video and audio communications. Devices, for example, may be placed adjacent to an external door or entryway. However, conventional devices may be unaesthetically appealing, have large form factors, and/or have suboptimal audio and/or video characteristics. Additionally, current devices may fail to accurately warn homeowners, businesses, and the like of imminent threats, such as detecting intruders or unwelcomed guests.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The detailed description is described with reference to the accompanying figures. In the figures, the left-most digit(s) of a reference number identifies the figure in which the reference number first appears. The use of the same reference numbers in different figures indicates similar or identical

components or features. The components, devices, and/or apparatuses depicted in the accompanying figures are not to scale and components within the figures may be depicted not to scale with each other.

(2) FIG. 1 illustrates a front perspective view of an example security camera device, according to examples of the present disclosure.

(3) FIG. 2 illustrates a rear perspective view of the security camera device of FIG. 1, according to examples of the present disclosure.

(4) FIG. 3 illustrates a front view of the security camera device of FIG. 1, according to examples of the present disclosure.

(5) FIG. 4 illustrates a rear view of the security camera device of FIG. 1, according to examples of the present disclosure.

(6) FIG. 5 illustrates a bottom view of the security camera device of FIG. 1, according to examples of the present disclosure.

(7) FIG. 6 illustrates a top view of the security camera device of FIG. 1, according to examples of the present disclosure.

(8) FIG. 7A illustrates a first side view of the security camera device of FIG. 1, according to examples of the present disclosure.

(9) FIG. 7B illustrates a second side view of the security camera device of FIG. 1, according to examples of the present disclosure.

(10) FIG. 8 illustrates a cross-sectional view of the security camera device of FIG. 1, taken along line A-A of FIG. 6, according to examples of the present disclosure.

(11) FIG. 9 illustrates a partially exploded view of the security camera device of FIG. 1, showing an example top (first) housing and an example bottom (second) housing of the security camera device, according to examples of the present disclosure.

(12) FIG. 10 illustrates a partially exploded view of the security camera device of FIG. 1, showing the top housing of FIG. 9 and an example mount configured to couple to the top housing, according to examples of the present disclosure.

(13) FIG. 11 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(14) FIG. 12 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(15) FIG. 13 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(16) FIG. 14 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(17) FIG. 15 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(18) FIG. 16 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(19) FIG. 17 illustrates example components disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(20) FIG. 18 illustrates an example battery housing disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(21) FIG. 19 illustrates an example sub-assembly disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(22) FIG. 20 illustrates an example midframe of the sub-assembly of FIG. 19, according to examples of the present disclosure.

(23) FIG. 21 illustrates an example printed circuit board assembly (PCBA) of the sub-assembly of FIG. 19, according to examples of the present disclosure.

(24) FIGS. 22A-22D illustrate an example passive infrared (PIR) assembly disposed within the top

housing of FIG. 9, according to examples of the present disclosure.

(25) FIGS. 23A-23D illustrate an example frame of the PIR assembly of FIGS. 22A-22D, according to examples of the present disclosure.

(26) FIGS. 24A and 24B illustrate example PIR sensor(s) and an example mount of the PIR assembly of FIGS. 22A-22D, according to examples of the present disclosure.

(27) FIGS. 25A and 25B illustrate the mount of FIGS. 24A and 24B for receiving the PIR sensor(s), according to examples of the present disclosure.

(28) FIGS. 26A and 26B illustrate an example lens through which the PIR assembly of FIGS. 22A-22D is configured to sense an environment, according to examples of the present disclosure.

(29) FIG. 27 illustrates a partially exploded view of the security camera device of FIG. 1, showing an example cap configured to couple to the top housing of FIG. 9, according to examples of the present disclosure.

(30) FIGS. 28A and 28B illustrate an example printed circuit board (PCB) disposed within the top housing of FIG. 9, according to examples of the present disclosure.

(31) FIG. 29 illustrates an example interior of the security camera device of FIG. 1, according to examples of the present disclosure.

(32) FIGS. 30A and 30B illustrate example batteries and contacts of the top housing of FIG. 9, according to examples of the present disclosure.

(33) FIGS. 31A-31C illustrate an example cover of the top housing of FIG. 9, according to examples of the present disclosure.

(34) FIGS. 32A-32C illustrate the bottom housing of FIG. 9, according to examples of the present disclosure.

(35) FIGS. 33A-33C illustrate an example loudspeaker of the bottom housing of FIG. 9, and coupling between the top housing and the bottom housing, according to examples of the present disclosure.

(36) FIGS. 34A-34D illustrate an example frame of the bottom housing of FIG. 9, according to examples of the present disclosure.

(37) FIGS. 35A-35C illustrate an example cover of the bottom housing of FIG. 9, according to examples of the present disclosure.

(38) FIG. 36 illustrates an example loudspeaker grill of the bottom housing of FIG. 9, according to examples of the present disclosure.

(39) FIG. 37 illustrates example computing components of the security camera device of FIG. 1, according to examples of the present disclosure.

DETAILED DESCRIPTION

(40) This application is directed, at least in part, to a security camera device (e.g., electronic device, audio/video (A/V) device, etc.) having improved image, audio, and video capturing characteristics. In some instances, the security camera device may include a first housing and a second housing that couple together. Additionally, a mount may couple to the first housing for disposing the security camera device on a surface (e.g., wall). The first housing may include various computing components, such as a camera, light emitting elements (e.g., light emitting diodes (LEDs)), RADAR, passive infrared (PIR) sensor(s), and so forth for detecting motion, recording audio and/or video, and so forth within an environment of the security camera device. The second housing may include a loudspeaker (e.g., mid-range speaker, tweeter speaker, or subwoofer speaker) for outputting sound within the environment. The first housing and the second housing may communicatively couple to one another via various connections (e.g., contacts) to enable operation of the security camera device.

(41) The security camera device may include a front, a back, a top, a bottom, and adjacent lateral sides. In some instances, the first housing includes the camera(s), the PIR sensor(s), and/or the RADAR oriented towards the front. In some instances, the camera(s) may be located closer to the top of the security camera device than the PIR sensor(s) and the RADAR, the PIR sensor(s) may be

located closer to the bottom of the security camera device than the camera(s) and the RADAR, and/or the RADAR may be located between the camera(s) and the PIR sensor(s). The camera(s), the PIR sensor(s), and/or the RADAR may serve to detect motion within an environment of the security camera device. For example, image processing on video data captured by the camera(s) may be used to detect persons (or other objects of interest). The PIR sensor(s) detect or receive mid-IR light emitted by persons (e.g., according to a temperature of the person). The RADAR may be used to determine polar coordinates (or a position) of persons. In some instances, the RADAR may enable the security camera device to focus on object detection and capabilities to operation in low power modes. In some instances, the camera(s), the PIR sensor(s), and/or the RADAR may be used in combination to detect persons, and in response, capture video and/or audio data. For example, upon mid-IR (e.g., motion) being detected via the PIR sensor(s), the camera(s) may begin recording video data.

(42) A PIR sensor may comprise, for example, two pyroelectric sensing elements. Each pyroelectric sensing element comprises a pyroelectric crystal. Each pyroelectric sensing element generates an electrical charge in response to heat. Radiation (e.g. infrared light) received at a surface of a pyroelectric sensing element generates heat, which in turn generates an electrical charge. Put another way, an absorbing layer of a pyroelectric sensing element transforms radiation flux change into a change in temperature and a pyroelectric component performs a thermal to electrical conversion. One or more low-noise and low leakage current field-effect transistors (e.g. JFET) or operational amplifiers are used to convert charge into a signal voltage.

(43) In some instances, a PIR sensor may comprise two pyroelectric sensing elements electrically coupled together with opposite polarization to produce an output. In this way, an equal change in temperature at both of the pyroelectric sensing elements will cancel out in the output signal, thus filtering out temperature changes in the environment. However, a change in temperature at only one of the pyroelectric sensing elements will result in an output signal that is positive or negative (depending on which pyroelectric sensing element experienced the change in temperature).

(44) A PIR sensor may include two slots, each providing an optical path to one of the pyroelectric sensing elements. A PIR sensor or electronic device may comprise one or more lenses configured to direct light received at the one or more lenses onto one of the pyroelectric sensing elements. A PIR sensor or electronic device may include one or more lenses configured to direct light received at a first portion of the one or more lenses (e.g. a left portion) onto a first of the pyroelectric sensing elements (e.g. a left sensing element), and to direct light received at a second portion of the one or more lenses (e.g. a right portion) onto a second of the pyroelectric sensing elements (e.g. a right sensing element). The one or more lenses may comprise one or more Fresnel lenses having one or more features configured to direct light. The pyroelectric elements may be positioned side by side and aligned along an axis (e.g. a horizontal axis or a vertical axis).

(45) A PIR sensor may be analog, with an analog signal output, or may be digital, with digital data output generated utilizing an analog-to-digital converter (ADC).

(46) An electronic device may include one or more PIR sensors that the electronic device uses to detect objects. Each PIR sensor may output a signal or sensor data, where the electronic device uses a characteristic determined using the signal or sensor data to determine whether the passive infrared sensor detected an object. The characteristic may include a voltage represented by the signal or sensor data, an amplitude of a wave generated or determined using the signal or sensor data, an angle of the wave generated using the signal or sensor data, and/or the like.

(47) For example, a first PIR sensor may have a first field-of-view (FOV) that extends a first distance from the electronic device. In some examples, the first FOV is created based on placing the first PIR sensor in a first direction and/or using one or more lenses (which may be a lens of the passive infrared sensor or which may be used in addition to or in replacement of a lens of the PIR sensor).

(48) The electronic device may include one or more lenses configured to direct light received at

various first portions of the one or more lenses onto a passive infrared sensor. The one or more lenses may comprise one or more Fresnel lenses having one or more features configured to direct light.

(49) In some instances, the PIR sensor(s) are part of a PIR assembly disposed within the first housing. The PIR assembly may include a mount, a PIR frame, a reflective surface (e.g., mirror), and the PIR sensor(s). The PIR sensor(s) may couple to the mount or be disposed on and/or within the mount. As such, the mount may receive the PIR sensor(s). In some instances, the mount may orient the PIR sensor(s) towards the sides, the top, and/or the front of the security camera device. The PIR sensor(s) and the mount may couple to the PIR frame. In some instances, the PIR frame includes two cavities (e.g., one for each PIR sensor), separated by a divider (e.g., wall), and through which the PIR sensor(s) are configured to receive IR radiation. In some instances, the mirror couples to the PIR frame and the PIR sensor(s) are oriented towards the mirror. The mirror is configured to reflect the IR signals in a direction towards the PIR sensor(s). The mirror may serve to increase a field of view of the PIR sensor(s). In some instances, the mirror is formed via two separate mirrors, or may be a unibody structure. In instances in which the device includes two mirrors, a first of the PIR sensor(s) may be oriented towards a first mirror, and a second of the PIR sensor(s) may be oriented towards a second mirror.

(50) The security camera device may include various LEDs for illuminating the environment and for enabling the camera(s) to capture video data and/or motion detection. In some instances, the LEDs may be disposed within the first housing. In some instances, the LEDs may include first LEDs (e.g., IR LEDs) arranged on the front of the security camera device, second LEDs (e.g., WLEDs) arranged at a first corner the security camera device, between the front and a first side, and/or third LEDs (e.g., WLEDs) arranged at a second corner of the security camera device, between the front and a second side. The first LEDs may enable the camera(s) to capture video data in low-light conditions (e.g., night time), while the second LEDs and/or the third LEDs emit light upon motion being detected. In some instances, the first LEDs are located vertically beneath the camera(s) and above the PIR sensor(s). Additionally, the first LEDs may be located horizontally between the second LEDs (at the first corner) and the third LEDs (at the second corner).

(51) The security camera device may include microphone(s) for capturing audio within the environment. In some instances, the microphone(s) are arranged on or across the front of the security camera device. In some instances, the microphone(s) may be located between the camera and the top of the device. As such, the microphone(s) may be located closer to the top of the security camera device than the camera(s). Additionally, the microphone(s) may include at least two microphones for noise cancellation, directionality, and so forth. In some instances, a first of the microphones is disposed proximate to the first corner, while a second of the microphones is disposed proximate the second corner (e.g., spaced apart across the front).

(52) Additionally, the front may include a status indicator (e.g., RGB LED) that indicates an operational state of the security camera device. For example, the status indicator may indicate whether the microphone(s) is/are recording audio, whether the camera(s) is/are capturing video, and so forth. In some instances, the status indicator is located between the first microphone and the second microphone, vertically above and/or aligned with camera(s). The front of the security camera device may also include an ambient light sensor that is used to transition the security camera device to night mode (e.g., low light conditions). The ambient light sensor may be interposed between the status indicator and the camera. In some instances, the microphone(s), status indicator, and/or ambient light sensor are disposed within the first housing.

(53) The top of the first housing may include a button, which in some instances, may be used to setup the security camera device. For example, as part of an out of box experience (OOBE), the button may be pressed to setup the security camera device. In some instances, the button is disposed through a cap of the first housing. In some instances, the button may be a mechanical button, resistive-type button, or a capacitive-type button. The cap may be located on the top of the

security camera device, and secured to the first housing via fasteners, snap-fits, press-fit, etc.

(54) The first housing may also include a battery frame (e.g., housing) that defines one or more battery receptacles. For example, the battery frame may include a body that defines a first battery receptacle and a second battery receptacle for receiving a first battery and a second battery, respectively. In some instances, the battery receptacles may be accessible via a bottom of the first housing, and by uncoupling the first housing and the second housing.

(55) In some instances, one or more printed circuit boards (PCBs) or printed circuit board assemblies (PCBAs) are disposed within the first housing. In some instances, a first PCBA may be disposed at or beneath the front of the security camera device. The first PCBA may include, or communicatively couple to, the camera(s), the PIR sensor(s), the RADAR, the first LEDs (IR LEDs), the microphone(s), and so forth. A PCB may be disposed at or beneath the cap of the security camera device. In some instances, the PCB may include network interfaces (e.g., antenna(s)) for wirelessly coupling the security camera device with one or more remote devices (e.g., mobile device, laptop, etc.) over one or more networks (e.g., Cellular, Wi-Fi, BLE, Bluetooth, etc.). The PCB may also receive power from the batteries of the security camera device. Additionally, the second LEDs and the third LEDs arranged at corners may be disposed on one or more PCBAs. For example, a second PCBA (or PCB) may include the second LEDs disposed at the first corner of the security camera device, while a third PCBA (or PCB) may include the third LEDs located at the second corner of the security camera device. The PCBAs and the PCB (as well as other computing components of the security camera device) may be communicatively coupled to one another via one or more flex circuits, wires, connectors, and so forth. For example, the PCBAs and the PCBs may be communicatively coupled for distributing power to components of the security camera device.

(56) The security camera device may include one or more windows or lenses disposed over the camera(s), PIR sensor(s), RADAR, and/or the LED(s). For example, a window may be disposed over the camera(s), a first lens may be disposed over the first LEDs and/or the RADAR, a second lens may be disposed over the PIR sensor(s), a third lens may be disposed over the second LEDs at the first corner, and/or a fourth lens may be disposed over the third LEDs at the second corner. In some instances, the window may be received within at least a portion of the first lens. For example, the window may reside within an opening of the first lens. In some instances, the windows and/or lenses may couple to a frame of the first housing, at the front and/or along the sides. For example, as the second LEDs and the third LEDs may be disposed at corners of the security camera device, the third lens and/or the fourth lens may at least partially wrap around the front and/or the sides of the security camera device. The windows and/or lenses may also conceal the camera(s), PIR sensor(s), RADAR, and/or the LED(s) from view. In some instances, the windows and/or lenses may include a material that is transparent or translucent to permit the PIR sensor(s) to detect motion and the cameras to record video, for example. Example materials may include glass and polycarbonate. The windows and/or lenses may also reduce glare (e.g., anti-glare coatings) to increase a quality of videos being captured.

(57) The second housing includes the loudspeaker for outputting sound into the environment. In some instances, the loudspeaker is arranged to output sound towards the front and/or the bottom of the security camera device. For example, the loudspeaker may be oriented towards the front and the bottom to project sound outward from the security camera device. The second housing may include a frame to which the loudspeaker couples for orienting the loudspeaker towards the front and the bottom. In some instances, a loudspeaker grill and/or other membranes are disposed over the loudspeaker.

(58) In some instances, the first housing and the second housing may couple together via a twist and lock mechanism. Attachment mechanisms of the first housing and the second housing may form the twist and lock mechanism. For example, the first housing may include first attachment mechanisms that engage with second attachment mechanism of the second housing. In some

instances, the first attachment mechanisms may angularly span around at least a portion of a perimeter, circumference, or periphery of the first housing, while the second attachment mechanisms may span around at least a portion of an opening, annulus, or receptacle of the second housing. The first attachment mechanisms and the second attachments mechanism may engage (e.g., interlock) via a rotation of the first housing and/or the second housing. For example, rotating the first housing and/or the second housing in a first direction (e.g., clockwise) may lock the first attachment mechanisms and the second attachment mechanisms together. When engaged, the first housing and the second housing may be securely coupled together. Rotating the first housing and/or the second housing in a second direction (e.g., counterclockwise) may unlock the first attachment mechanisms and the second attachment mechanisms such that the first housing and the second housing may be uncoupled. The first attachment mechanisms may resemble tabs, hooks, protrusions, keys, keyways, slots, or other male/female connectors that are complimentary to engage with the second attachment mechanisms.

(59) The first housing may also include first contacts that engage with second contacts of the second housing. When the first housing and the second housing are coupled together, the first contacts and the second contacts may engage. In some instances, the engagement between the first contacts and the second contacts transfer power, data, and so forth between computing components in the first housing and computing components in the second housing. For example, power may be transferred between the first housing and the second housing via an engagement between the first contacts and the second contacts. Additionally, sound (e.g., audio data/audio signal) that is output via the loudspeaker may be transferred via the first contacts and the second contacts. In some instances, the first contacts represent prongs, pins, tabs, etc. and the second contacts represent races, strips, pads, etc. that receive the prongs.

(60) The mount may include a ball and socket joint that couples to the first housing. In some instances, an arm with a ball may extend from the back of the first housing. The arm extends the ball away from the back of the first housing. The ball may be received within a socket of the mount, and a clamp of the socket (e.g., fastener) may couple the ball and socket together. The ball and socket joint may enable an orientation of the security camera device to be changed. In some instances, the mount may be coupled (e.g., fastened) to surfaces (e.g., walls, roof eave, etc.). Additionally, the back may include a port, such as a USB-C port. However, additional or alternative ports are envisioned (e.g., auxiliary, ethernet, etc.) and/or the port(s) may be located elsewhere than the back. In some instances, the port may be used for charging batteries of the security camera device (e.g., power), connecting the security camera device to networks, connecting the security camera device to solar panels, and so forth.

(61) The first housing and the second housing may include covers that form an exterior surface of the security camera device. For example, the first housing may include a first cover and the second housing may include a second cover. The first cover and the second cover may include cutouts, apertures, recesses, and the like for accommodating or otherwise receiving the windows, lenses, loudspeaker grill, and so forth, respectively. By way of example, the third lens and the fourth lens may be received within cutouts of the first cover. In some instances, the lenses, windows, and covers may couple to create the uniform exterior surface of the security camera device.

(62) The first housing and the second housing may include various frames, mounts, and brackets for receiving components of the security camera device, or to which components of the security camera device couple. For example, a frame of the first housing may include a cutout through which the camera is disposed, and/or to which PCBAs and/or PCBs couple. In some instances, the covers of the first housing and the second housing couple to the frames. The frames may also include various alignment mechanisms, such as tabs, pins, slits, and so forth for aligning components within the security camera device. For example, prongs extending from a frame may be disposed through openings on the PCBAs and/or the PCBs for aligning the PCBAs and/or the PCBs on the frame (and within the security camera device). Various seals (e.g., O-rings, gaskets,

etc.) may be included to environmentally seal the security camera device.

(63) In some instances, heat dissipating elements are included to disperse heat generated by components of the security camera device. By way of example and not limitation, the camera(s), LEDs, power supply, network interfaces, and so forth generate heat during use. Without effectively dispersing or dissipating this heat, the internal components, such as a camera, may be adversely effected and become uncappable of performing its intended function. To efficiently dissipate heat generated by the components, heat dissipating elements may be included within the first housing and/or the second housing to transmit heat away from generating sources toward an exterior of the first housing and/or the second housing, and/or to uniformly distribute the heat over the surface area of the security camera device.

(64) Components of the security camera device may be manufactured using any suitable manufacturing technique, such as blow molding, injection molding, stamping, and so forth. Additionally, suitable materials include metals, composites, plastics, and/or any combination thereof.

(65) The present disclosure provides an overall understanding of the principles of the structure, function, device, and system disclosed herein. One or more examples of the present disclosure are illustrated in the accompanying drawings. Those of ordinary skill in the art will understand that the devices and/or the systems specifically described herein and illustrated in the accompanying drawings are non-limiting embodiments. The features illustrated or described in connection with one embodiment may be combined with the features of other embodiments. Such modifications and variations are intended to be included within the scope of the appended claims.

(66) FIG. 1 illustrates a front perspective view of an example security camera device **100**, according to examples of the present disclosure. In some instances, the security camera device **100** may represent an electronic device, audio/video (A/V) device, and so forth. The security camera device **100** may include a first housing **102** (e.g., top housing) that couples to a second housing **104** (e.g., bottom housing). As shown, the first housing **102** may reside vertically above the second housing **104**, such that the second housing **104** may couple to a bottom of the first housing **102**. When coupled together, the first housing **102** and the second housing **104** may form an exterior surface of the security camera device **100**. Additionally, a mount **106** may couple to the first housing **102**. The mount **106** may be used to secure the security camera device **100** to surfaces (e.g., wall). For example, fasteners may be disposed through the mount **106** for securing the security camera device **100** to the surfaces.

(67) As will be explained herein, the first housing **102** and the second housing **104** may couple together via a twist and lock mechanism. For example, rotating the first housing **102** and/or the second housing **104** in a first direction may couple the first housing **102** and the second housing **104** together, while rotating the first housing **102** and/or the second housing **104** in a second direction (opposite the first direction) may uncouple the first housing **102** and the second housing **104**.

(68) The security camera device **100** includes a front **108**, a back **110**, a top **112**, a bottom **114**, and adjacent lateral sides, such as a first side **116** and a second side **118**. The first housing **102** may include a first cover **120** and a cap **122**. The first cover **120** may be disposed along the front **108**, the back **110**, the first side **116**, and/or the second side **118** of the security camera device **100**. The cap **122**, meanwhile, may couple to a top of the first cover **120**, to form the top **112** of the security camera device **100**. Seals or gaskets may be disposed at an interfaces between the cap **122** and the first cover **120**. The second housing **104** may include a second cover **124** disposed along the front **108**, the back **110**, the bottom **114**, the first side **116**, and/or the second side **118** of the security camera device **100**.

(69) The front **108** may include a window **126**, a first lens **128**, a second lens **130**, a third lens **132**, and a fourth lens **134**. As shown, the window **126**, the first lens **128**, and the second lens **130** may reside on the front **108** of the security camera device **100**, while the third lens **132** may reside on

the front **108** and/or the first side **116** of the security camera device **100** (e.g., first corner) and the fourth lens **134** may reside on the front **108** and/or the second side **118** of the security camera device **100** (e.g., second corner). Additionally, the window **126** may reside within the first lens **128** (e.g., an opening of the first lens **128**). The second lens **130** resides vertically beneath the window **126** and the first lens **128**.

(70) In some instances, the window **126**, the first lens **128**, the second lens **130**, the third lens **132**, and the fourth lens **134** are part of the first housing **102**. The window **126**, the first lens **128**, the second lens **130**, the third lens **132**, and the fourth lens **134** may also couple to the first cover **120**. As will be explained herein, the first cover **120** may include cutouts, pockets, and the like for receiving the window **126**, the first lens **128**, the second lens **130**, the third lens **132**, and/or the fourth lens **134**. As will also be explained herein, a camera of the security camera device **100** may reside within the first housing **102** and beneath the window **126**, a RADAR and/or IR LEDs of the security camera device **100** may reside within the first housing **102** and beneath the first lens **128**, PIR sensor(s) of the security camera device **100** may reside within the first housing **102** and beneath the second lens **130**, LEDs of the security camera device **100** may reside within the first housing **102** and beneath the third lens **132**, and/or LEDs of the security camera device **100** may reside within the first housing **102** and beneath the fourth lens **134**.

(71) In some instances, the window **126**, the first lens **128**, the second lens **130**, the third lens **132**, and/or the fourth lens **134** may include a material that is transparent or translucent to permit the camera, PIR sensor(s), and LEDs to detect motion, emit light, and so forth, respectively. Example materials may include glass and polycarbonate.

(72) The top **112** includes a button **136**, which in some instances, is part of the cap **122** and/or is disposed through an opening in the cap **122**. The button **136** may correspond to a setup button, a power button, a wireless connectivity button, a mute button, a volume button, a sync button, or any other type of button or control. The button **136** may be mechanical (e.g., having physically movable components) and/or electronic (e.g., capacitive sensors, optical sensors, touch screen, or the like).

(73) The second housing **104**, as explained herein, may include a loudspeaker oriented to emit sound in a direction away from the security camera device **100**. A loudspeaker grill **138** (e.g., speaker mesh) may couple to the second cover **124** to permit sound generated by the loudspeaker to exit the security camera device **100**.

(74) FIG. 2 illustrates a rear perspective view of the security camera device **100**, according to examples of the present disclosure. The security camera device **100** includes the first housing **102** having the third lens **132** disposed on the front **108** and the first side **116** and the button **136** located on the top **112**. Additionally, the first housing **102** includes an arm **200** that extends from the back **110** of the security camera device **100** or the first cover **120**. The arm **200** may be received by (or within) the mount **106**. For example, the mount **106** may include a base **202** for securing the security camera device **100** to a surface (via fasteners), a socket **204**, and a clamp **206**. The socket **204** may represent a receptacle for receiving the arm **200** (e.g., a ball of the arm **200**). The clamp **206** may be tightened and loosened such that an orientation of the security camera device **100** may be adjusted.

(75) The back **110** is also shown including a port **208**, which in some instances, may represent a USB-C port. The port **208** may accept a cable for receiving power to power components of the security camera device **100**, for recharging batteries of the security camera device **100**, for connecting with one or more networks (e.g., Ethernet cable), for connecting the security camera device **100** to a solar panel, and so forth.

(76) The second housing **104** is shown including the loudspeaker grill **138** disposed on the bottom **114** of the security camera device **100**, coupled to the second cover **124**.

(77) FIG. 3 illustrates a front view of the security camera device **100**, according to examples of the present disclosure. The first housing **102** is shown including the window **126**, the first lens **128**, the

second lens **130**, the third lens **132**, and the fourth lens **134**. The first lens **128**, the second lens **130**, the third lens **132**, and the fourth lens **134** may be disposed with receptacles of the first cover **120**. The first lens **128** is shown including an opening **300** (e.g., circular opening) through which the window **126** is disposed, such that the window **126** may reside within the first lens **128**. The second lens **130** is shown located vertically below the window **126** and the first lens **128**. The window **126**, the first lens **128**, and the second lens **130** also reside between the third lens **132** and the fourth lens **134** (e.g., horizontally). In some instances, the window **126** may be substantially circular shaped, the first lens **128** may be substantially rectangular shaped, the second lens **130** may be substantially rectangular shaped, the third lens **132** may be substantially rectangular shaped, and/or the fourth lens **134** may be substantially rectangular shaped. In some instances, a top of the first lens **128** may be aligned with a top of the third lens **132** and/or the fourth lens **134**, and/or a bottom of the first lens **128** may be aligned with a bottom of the third lens **132** and/or the fourth lens **134**.

(78) The first lens **128** may define a first microphone port **302(1)** and a second microphone port **302(2)**. In some instances, the microphone ports **302** may be located closer to the top **112** than the window **126**. In some instances, the first microphone port **302(1)** is located closer to the first side **116** of the security camera device **100**, and the second microphone port **302(2)** is located closer to the second side **118** of the security camera device **100**. As will be explained herein, microphone(s) of the security camera device **100** may reside beneath (Z-direction) the microphone ports **302** such that sound may reach the microphones. That is, the microphone(s) may receive sound, for instance, via the microphone ports **302**. In some instances, the microphone ports **302** may be aligned with the microphones, or vice versa. Additionally, the microphones and/or the microphone ports **302** may be encased with foam that acoustically seals the microphones to minimize sound received and/or generated via other portions of the security camera device **100**. For example, the foam may isolate the microphones from one or more of the loudspeaker(s) to minimize an intensity of audio received from the loudspeaker(s).

(79) The first lens **128** may also include an area **304** through which light is emitted via a status indicator. For example, a status indicator may reside beneath the first lens **128**, and emit light through the area **304** of the first lens **128**. In some instances, the status indicator (e.g., RGB LED) indicates an operational state of the security camera device **100**. As shown, the area **304** of the first lens **128** may be located between the microphone ports **302**, and may be disposed vertically above the window **126**. In some instances, the area **304** may be an opening, or a portion of the first lens **128** that is transparent to light emitted by the status indicator. In some instances, the status indicator is centered between the microphone ports **302**.

(80) The top **112** includes the button **136**, which in some instances, is proud of the top **112** (e.g., extends beyond an exterior surface of the top **112**). In some instances, the button **136**, the area **304**, and the window **126** may be centrally aligned (e.g., in the X-direction). As shown, the first housing **102** (or the first cover **120**), may be substantially rectangular shaped, while the second housing **104** (or the second cover **124**) may include a taper. For example, in a direction between the top **112** and the bottom **114**, the exterior surface may taper.

(81) FIG. 4 illustrates a rear view of the security camera device **100**, according to examples of the present disclosure. In some instances, the port **208** and the arm **20** are centrally located (e.g., in the X-direction) on the back **110**. In doing so, the mount **106** may be aligned with the security camera device **100**.

(82) In some instances, and as noted above in FIG. 4, the security camera device **100** may taper between the top **112** and the bottom **114**. For example, security camera device **100** (or the first housing **102** or the first cover **120**) may include a first width **400** (e.g., in the X-direction) proximate to the top **112**, and the second housing **104** or the second cover **124** has a second width **402** (e.g., in the X-direction) proximate to the bottom **114**. The first width **400** may be greater than the second width **402**. In some instances, the first housing **102** may include the first width **400** along a height (e.g., in the Y-direction) of the first housing **102**. In some instances, the second

housing **104** may include a variable width along a height of the second housing **104** (e.g., the first width **400**, the second width **402**, a width of the loudspeaker grill **138**, and so forth).

(83) The mount **106** may include various holes, channels, and so forth for mounting the security camera device **100** to a surface. In some instances, a rubber foot, for example, may be disposed on the mount **106** to reduce vibrations being imparted to the security camera device **100** and/or to secure the mount **106** to the surface (e.g., increased friction).

(84) FIG. 5 illustrates a bottom view of the security camera device **100**, according to examples of the present disclosure. The bottom **114** is shown including the loudspeaker grill **138** through which sound output by the loudspeaker(s) passes. The arm **200** is also shown extending from the back **110** of the first housing **102** for engaging with the mount **106**.

(85) As noted above, the first housing **102** and the second housing **104** may couple together via a twist and lock mechanism. In some instances, a fastener **500** may also be used to couple to the first housing **102** and the second housing **104** together. For example, after the first housing **102** and the second housing **104** are coupled together via the twist and lock mechanism, the fastener **500** may be used to further secure the first housing **102** and the second housing **104** together. This may, for example, prevent an uncoupling of the first housing **102** and the second housing **104**. That is, the first housing **102** and the second housing **104** may be restricted from rotating when the fastener **500** is tightened (e.g., thereby coupling the first housing **102** and the second housing **104**).

(86) FIG. 6 illustrates a top view of the security camera device **100**, according to examples of the present disclosure. The security camera device **100** includes the front **108**, the back **110**, the first side **116**, and the second side **118**. The arm **200** is shown extending from the back **110** for coupling to the mount **106**. The arm **200** includes a ball **600** that is received within the socket **204** of the mount **106**, and the clamp **206** secures the ball **600** and the socket **204** together.

(87) A line A-A is shown extending through the security camera device **100**, which is used to illustrate a cross-sectional view of the security camera device in FIG. 8.

(88) FIGS. 7A and 7B illustrate side views of the security camera device **100**, according to examples of the present disclosure. FIG. 7A illustrates the first side **116** of the security camera device **100**, while FIG. 7B illustrates the second side **118** of the security camera device **100**. The third lens **132** may extend from the front **108** onto the first side **116**, while the fourth lens **134** may extend from the front **108** onto the second side **118**.

(89) The loudspeaker may be oriented in a direction to output sound out and away from the security camera device **100**. Orienting the loudspeaker in this manner may project sound outwards, in front of the security camera device **100** and/or to the sides of the security camera device **100**. In other words, the loudspeaker may be oriented away from the back **110** of the security camera device **100**.

(90) FIG. 8 illustrates a cross-sectional view of the security camera device **100**, taken along line A-A of FIG. 6, according to examples of the present disclosure. The first cover **120** and the second cover **124** are shown being disposed around components of the first housing **102** and the second housing **104**, respectively, such that components of the first housing **102** and the second housing **104** reside within the first cover **120** and the second cover **124**, respectively. Additionally, the cap **122** is shown being coupled to the first cover **120**, and the loudspeaker grill **138** is shown being coupled to the second cover **124**.

(91) The security camera device **100** is shown including a battery housing **800**, a camera **802**, PIR sensor(s) **804**, and a RADAR sensor **806**. The battery housing **800**, the camera **802**, the PIR sensor(s) **804**, and the RADAR sensor **806** are shown being disposed within the first housing **102**. The camera **802** and the PIR sensor(s) **804** may be oriented towards the front of the security camera device **100** (e.g., in a first direction). Additionally, the camera **802** is disposed beneath (e.g., Z-direction) the window **126**, while the RADAR sensor **806** may be disposed beneath (e.g., Z-direction) the first lens **128**. The PIR sensor(s) **804** may be disposed beneath the second lens **130**. As will be explained herein, the battery housing **800** may include receptacles for receiving batteries (e.g., two batteries) of the security camera device **100**.

(92) The security camera device **100** may include various PCBAs and/or PCBs for controlling an operation of the security camera device **100** and/or to enable function of the security camera device **100**. Any number of PCBAs and/or PCBs (or like processors, memory, modules, controllers, etc.) may be included. In some instances, the PCBAs and/or PCBs may be disposed in the first housing **102** and/or the second housing **104**. For example, the first housing **102** may include a PCB **808** and a PCBA **810**. The PCB **808** may be disposed beneath the cap **122** (e.g., in the Y-direction), vertically above the battery housing **800**. The PCB **808** may include or communicatively couple to the button **136**. Additionally, as will be explained herein, the PCB **808** may include communicatively couple to network interfaces (e.g., antenna(s)) of the security camera device **100**. The PCB **808** may also communicatively couple to the batteries within the battery housing **800**, as well as the port **208**.

(93) The PCBA **810** may be disposed on the front **108** of the security camera device **100**, between the battery housing **800** and the front **108**. In some instances, the camera **802**, the PIR sensor(s) **804**, the RADAR sensor **806**, as well as other components, may couple to, be disposed on, or communicatively couple to the PCBA **810**. Various connections (e.g., flex circuits) may communicatively couple the PCB **808**, the PCBA **810**, and other computing components of the security camera device **100**. Additionally, the security camera device **100** may include a contact connection **812** between the first housing **102** and the second housing **104** for communicatively coupling computing components in the first housing **102** with computing components in the second housing **104**. For example, power may be routed from the first housing **102** and the second housing **104**, via the contact connection **812**, for transmitting power to a loudspeaker **814**. As will be explained herein, the contact connection **812** may include first contacts of the first housing **102** that mate or otherwise engage with second contacts of the second housing **104**. The first contacts and the second contacts may be brought into contact when the first housing **102** and the second housing **104** couple together, and may be brought out of contact when the first housing **102** and the second housing **104** uncouple. In some instances, the PCBA **810** may couple to a front of the battery housing **800**, while the PCB **808** may couple to a top of the battery housing **800**.

(94) The loudspeaker **814** may reside beneath the loudspeaker grill **138**, and oriented in a direction away from the back **110** of the security camera device **100**, so as to output sound in front and to the sides of the security camera device **100**. In some instances, an axis **816** (vertical axis/plane) may be disposed orthogonally through a center of the security camera device **100**, and the loudspeaker **814** may be oriented relative to the axis **816**. In some instances, the loudspeaker **814** is oriented at an angle @ relative to the axis **816**. In some instances, the angle @ may be between approximately ten degrees and forty-five degrees. As such, the loudspeaker **814** may be oriented towards the front and away from the axis **816** (or a central vertical plan through the security camera device **100**).

(95) The arm **200** is also shown extending from the back **110** of the security camera device **100** (e.g., from the first cover **120**), and includes the ball **600** for being received by the mount **106**.

(96) FIG. 9 illustrates a partially exploded view of the security camera device **100**, showing the first housing **102** and the second housing **104** being separated (e.g., in the Y-direction), according to examples of the present disclosure.

(97) The first housing **102**, or the first cover **120**, may include first attachment mechanisms **900** that engage with second attachment mechanisms **902** of the second housing **104**. In some instances, the second attachment mechanisms **902** may be formed within a frame **904** of the second housing **104** to which the second cover **124** couples. In some instances, the first attachment mechanisms **900** may resemble tabs, hooks, protrusions, keys, keyways, slots, or other male/female connectors that are complimentary to engage with the second attachment mechanisms **902** (e.g., tabs, hooks, protrusions, keys, keyways, slots, or other male/female connectors).

(98) The first attachment mechanisms **900** may couple with the second attachment mechanisms **902** when the first housing **102** and/or the second housing **104** is rotated. That is, the first attachment mechanisms **900** may engage with and secure to the second attachment mechanisms **902** when the

first housing **102** and/or the second housing **104** are rotated into contact with one another. As an example, the first attachment mechanisms **900** may include a keyway, and the second attachment mechanisms **902** may include a hook or a tab that is configured to engage with the keyway when rotated. In some instances, given that the first housing **102** may couple to a surface (via the mount **106**), the second housing **104** may be more easily rotated to couple the first housing **102** and the second housing **104** together. In such instances, the second housing **104** may be rotated in a first direction (e.g., clockwise about the Y-axis) to engage the first attachment mechanisms **900** and the second attachment mechanisms **902**, while the second housing **104** may be rotated in a second direction (e.g., counterclockwise about the Y-axis) to disengage the first attachment mechanisms **900** and the second attachment mechanisms **902**. Additionally, in some instances, when engaged, the fastener **500** may be used to additionally couple the first housing **102** and the second housing **104**.

(99) In some instances, the first attachment mechanisms **900** may be formed on and/or within an exterior surface of the first cover **120**. The first attachment mechanisms **900** may be disposed annularly around a perimeter **908** at a bottom end of the first cover **120**. The second attachment mechanisms **902** may be formed within an annulus **906** of the frame **904**. In some instances, the second attachment mechanisms **902** may be disposed annularly around the annulus **906**. The perimeter **908** of the first cover **120** is insertable into the annulus **906** of the frame **904**, or stated alternatively, the annulus **906** is configured to be disposed over the perimeter **908** of the first cover **120** for engaging with the first attachment mechanisms **900**.

(100) Any number of the first attachment mechanisms **900** and the second attachment mechanisms **902** may be included. For example, the security camera device **100** may include two of the first attachment mechanisms **900** that engage with two of the second attachment mechanisms **902**. However, more than or less than two of the first attachment mechanisms **900** and the second attachment mechanisms **902** may be included.

(101) FIG. **10** illustrates a partially exploded view of the security camera device **100**, showing the mount **106** being separated from the first housing **102**, according to examples of the present disclosure. In some instances, the arm **200** and the ball **600** may be components of a plate **1000** that couples to the back **110** of the first housing **102** (or the first cover **120**). For example, the arm **200** may extend from the plate **1000** for engaging with the mount **106**. In some instances, the plate **1000** may couple to the first cover **120** via fasteners, snap-fits, pressure fits, and/or adhesives. The plate **1000** may also include a passage **1002** for accommodating the port **208**. The first attachment mechanisms **900** are also shown disposed on the first cover **120**.

(102) FIG. **11** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. The window **126**, the first lens **128**, the second lens **130**, the third lens **132**, and the fourth lens **134** are removed to illustrate components of first housing **102**.

(103) Removing the window **126**, the first lens **128**, the second lens **130**, the third lens **132**, and the fourth lens **134** exposes components of the first housing **102**. For example, the first housing **102** may include the RADAR sensor **806**, a RADAR PCB **1100**, an IR LED PCB **1102** having IR LEDs **1108**, a first LED PCB **1104**, and a second LED PCB **1106**. The RADAR PCB **1100**, the IR LED PCB **1102**, the first LED PCB **1104**, and the second LED PCB may communicatively couple to the PCB **808** and/or the PCBA **810**, as well as additional computing components of the security camera device **100**.

(104) The RADAR sensor **806** may be mounted on the RADAR PCB **1100**. In some instances, the RADAR PCB **1100** is disposed in front of the IR LED PCB **1102** (e.g., towards the front **108**). In some instances, the IR LEDs **1108** may include a first IR LED **1108(1)**, a second IR LED **1108(2)**, a third IR LED **1108(3)**, and a fourth IR LED **1108(4)**. In some instances, the IR LEDs **1108** may be disposed at corners of the IR LED PCB **1102**. The IR LEDs **1108** are oriented to emit IR light through the first lens **128**, outwards from the front **108** of the security camera device **100**. The IR LED PCB **1102** may include IR sensor(s) configured to detect the IR light reflected from objects

within an environment of the security camera device **100**. Such reflections may be used to detection motion around the security camera device **100**. The first IR LED **1108(1)** and the second IR LED **1108(2)** may be disposed to a first side of the RADAR sensor **806** and/or the RADAR PCB **1100**, and the third IR LED **1108(3)** and the fourth IR LED **1108(4)** may be disposed to a second side of the RADAR sensor **806** and/or the RADAR PCB **1100**.

(105) As shown and described in FIG. **12**, the first LED PCB **1104** may include one or more white LEDs (e.g., four) and the second LED PCB **1106** may include one or more white LEDs (e.g., four). The white LEDs may emit visible light. The first LED PCB **1104** may reside beneath the third lens **132** and the second LED PCB **1106** may reside beneath the fourth lens **134**. As such, the LEDs on the first LED PCB **1104** may emit light through the third lens **132**, and the LEDs on second LED PCB **1106** may emit light through the fourth lens **134**.

(106) The first housing **102** may also include a PIR assembly **1110**, which includes the PIR sensor(s) **804**, such as a first PIR sensor **804(1)** and a second PIR sensor **804(2)**. As shown, the PIR sensor(s) **804** are located vertically below the RADAR sensor **806**, the RADAR PCB **1100**, and IR LEDs **1108**. The PIR sensor(s) **804** are oriented to sense motion in front of the security camera device **100**, for example, via sensing mid-IR light emitted by persons. Additional details of the PIR assembly **1110** are discussed herein with regard to FIGS. **22A-26B**.

(107) FIG. **12** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. The first cover **120** is removed to illustrate components of first housing **102**. In some instances, the first LED PCB **1104** may include first LEDs **1200** (e.g., four), and/or the second LED PCB **1106** may include second LEDs **1202** (e.g., four). In some instances, the first LEDs **1200** and the second LEDs **1202** may be arranged in a vertical manner on the front **108** of the security camera device **100**, behind the third lens **132** and the fourth lens **134**, respectively. In some instances, the first LEDs **1200** and the second LEDs **1202** may be equidistantly spaced apart along a length of the first LED PCB **1104** and the second LED PCB **1106**. The first LED PCB **1104** and the second LED PCB **1106** may be angled outwards so as to output light in front of and to the sides of the security camera device **100**, respectively.

(108) The first housing **102** includes first foam **1204** for acoustically sealing a first microphone of the security camera device **100**, and second foam **1206** for acoustically sealing a second microphone of the security camera device **100**. The first foam **1204** and the second foam **1206** may include ports (e.g., openings) through which sound travels to reach the microphones, respectively. The camera **802** resides above the RADAR sensor **806**, horizontally between the first LED PCB **1104** and the second LED PCB **1106**. In some instances, the camera **802** may be centrally aligned with the RADAR sensor **806**, the RADAR PCB **1100**, and the IR LED PCB **1102**.

(109) As shown, the camera **802**, the IR LEDs **1108**, and/or the RADAR sensor **806** may be oriented in the same direction (e.g., towards the front). The first LEDs **1200** and the second LEDs **1202** may be oriented in towards the front and/or the sides, respectively.

(110) FIG. **13** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. From FIG. **12**, the first LED PCB **1104**, the second LED PCB **1106**, the RADAR sensor **806**, the RADAR PCB **1100**, the first foam **1204**, and the second foam **1206** are removed to illustrate components of first housing **102**.

(111) Removing the first foam **1204** and the second foam **1206** exposes a microphone flexible printed circuit (FPC) **1300**. The microphone FPC **1300** may communicatively couple to microphones of the security camera device **100**. The microphone FPC **1300** communicatively couples to the PCB **808** and the PCBA **810**. A status indicator **1302** (e.g., LED) is shown being located above the camera **802**, as well as an ambient light sensor **1304**. The status indicator **1302** may output information associated with an operation state of the security camera device **100**, such as whether the microphone(s) are recording audio, whether the camera **802** is capturing video, and so forth. The ambient light sensor **1304** may be used to determine lighting conditions of the environment of the security camera device **100**, for example, when switching between daytime and

nighttime modes (or light and dark lighting conditions). In some instances, the camera **802**, the status indicator **1302**, and the ambient light sensor **1304** may be centrally aligned with one another (e.g., in the X-direction).

(112) The first LED PCB **1104** and the second LED PCB **1106**, as well as other components of the first housing **102** such as the IR LED PCB **1102**, may couple to a heat dissipating element **1306**. In some instances, the heat dissipating element **1306** is disposed across the front **108**, the first side **116**, and/or the second side **118** of the security camera device **100**. The heat dissipating element **1306** may assist in dispersing heat away from the first LED PCB **1104** and the second LED PCB **1106** (or the LEDs disposed thereon). As will be explained herein, the heat dissipating element **1306** may couple to a midframe of the first housing **102**. The heat dissipating element **1306** also provides openings for routing power and/or signals to the first LED PCB **1104** (and the first LEDs **1200**) and the second LED PCB **1106** (and the second LEDs **1202**). For example, a first wire **1308** may couple to the first LED PCB **1104**, and a second wire **1310** may couple to the second LED PCB **1106**. In some instances, an opposite end of the first wire **1308** and the second wire **1310** not coupled to the first LED PCB **1104** and second LED PCB **1106**, respectively, may couple to the PCB **808**.

(113) The IR LED PCB **1102** is shown including a plurality of RADAR pad(s) **1312**. The RADAR pad(s) **1312** may represent pads disposed about the IR LED PCB **1102**. The RADAR pad(s) **1312** may receive RADAR signals emitted by the RADAR sensor **806** for determining coordinates and/or a position of motion within the environment. Although a particular number of the RADAR pad(s) **1312** are shown, any number of RADAR pad(s) **1312** may be included. In some instances, the RADAR pad(s) **1312** may be arranged around an opening **1314** of the IR LED PCB **1102**. For example, a flex circuit may route through the opening **1314** to communicatively couple to the RADAR sensor **806** (and the RADAR sensor **806**).

(114) FIG. **14** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. From FIG. **13**, the IR LED PCB **1102** (including the IR LEDs **1108**), the microphone FPC **1300**, the camera **802**, and the heat dissipating element **1306** are removed to illustrate components of first housing **102**.

(115) The security camera device **100** includes microphones **1400**, such as a first microphone **1400(1)** and a second microphone **1400(2)**. The first microphone **1400(1)** may be aligned with the first microphone port **302(1)** and the second microphone **1400(2)** may be aligned with the second microphone port **302(2)**. In doing so, the microphone ports **302** assist in transferring or directing sound that is external to the security camera device **100** to the microphones **1400** located within the security camera device **100**. That is, the microphones **1400** may receive audio, for instance, user speech, via the microphone ports **302**.

(116) The status indicator **1302** may be located between (e.g., interposed) the first microphone **1400(1)** and the second microphone **1400(2)**, vertically above and/or aligned with camera **802** and/or the ambient light sensor **1304**. In some instances, the first microphone **1400(1)** and the second microphone **1400(2)** may be aligned (e.g., horizontally) with the status indicator **1302**.

(117) The first housing **102** may include a midframe **1402** to which components of the first housing **102** couple. For example, the heat dissipating element **1306** may couple to the midframe **1402**. In some instances, the midframe **1402** may include prongs, tabs, receptacles, slits, etc. that engage with corresponding features of the components that couple to the midframe **1402** for orienting and/or aligning the components within the security camera device **100**. By way of example, prongs may extend from the midframe **1402** and which are disposed through the first LED PCB **1104** and the second LED PCB **1106**, respectively. Such disposition may orient the first LED PCB **1104** and the second LED PCB **1106**. In other words, the midframe **1402** may orient the first LEDs **1200** and the second LEDs **1202** towards the sides and front **108** of the security camera device **100**. Prongs may also similarly extend through the heat dissipating element **1306**. In some instances, the microphones **1400** are mounted to the midframe **1402**.

(118) The midframe **1402** is shown including an aperture **1404**. In some instances, the camera **802** is at least partially disposed through the aperture **1404**, or through which the camera **802** is configured to capture video data of the environment. A flex circuit **1406** is shown being disposed through the midframe **1402**, which in some instances, may communicatively couple the RADAR PCB **1100** (and the RADAR sensor **806**) with the PCBA **810**.

(119) FIG. **15** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. From FIG. **14**, the microphones **1400**, the ambient light sensor **1304**, the status indicator **1302**, and the midframe **1402** are removed to illustrate components of first housing **102**.

(120) The PCBA **810** is shown coupled to a battery housing **800**, which as explained herein, receives or otherwise houses batteries of the security camera device **100**. For example, the PCBA **810** may couple to a front of the battery housing **800**, opposite a rear of the battery housing **800**. The PCB **808** may couple to a top of the battery housing **800**, opposite a bottom of the battery housing **800** in which the batteries are insertable into the battery housing **800**. In some instances, the heat dissipating element **1306** may be disposed above the battery housing **800** and/or along one or more sides of the battery housing **800**. The heat dissipating element **1306** may dissipate heat generated by components of the PCB **808**.

(121) An antenna **1500** may reside to a side of the battery housing **800**, opposite a side in which a portion of the heat dissipating element **1306** is disposed. In some instances, the antenna **1500** may correspond to a long-range antenna. The antenna **1500** may be disposed along an inside surface of the first cover **120** when the security camera device **100** is assembled.

(122) The flex circuit **1406** may communicatively couple the PCBA **810** to the RADAR sensor **806**. The flex circuit **1406** may route through the opening **1314** for coupling to the RADAR sensor **806**. Additionally, a flex circuit **1502** may communicatively couple the PCB **808** to the PCBA **810**. The first wire **1308** and the second wire **1310** communicatively couple the PCB **808** to the first LED PCB **1104** and the second LED PCB **1106**, respectively. The PCBA **810** may include various shields **1504** disposed over computing components to reduce magnetic, conductive, and/or other interferences. Further, the PIR assembly **1110** may couple to the PCBA **810**.

(123) FIG. **16** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. From FIG. **15**, the cap **122**, the first wire **1308**, the second wire **1310**, and portions of the PIR assembly **1110**, such as a frame of the PIR assembly **1110**, are removed to illustrate components of first housing **102**. The PCB **808** resides vertically beneath the cap **122**, above the heat dissipating element **1306** and a top of the battery housing **800**. The flex circuit **1502** routes between the PCB **808** and the PCBA **810**. The first PIR sensor **804(1)** and the second PIR sensor **804(2)** couple to PCBA **810** for being oriented outwards from the front **108** of the security camera device **100**.

(124) FIG. **17** illustrates a frontal view of the first housing **102**, according to examples of the present disclosure. From FIG. **16**, the PCBA **810** is removed to illustrate components of first housing **102**.

(125) The PCB **808** may be spaced apart from a top end **1700** of the battery housing **800**. In some instances, the top end **1700** of the battery housing **800** includes posts **1702** to which the PCB **808** couples. For example, the posts **1702** may end from the battery housing **800**, and the PCB **808** and/or the heat dissipating element **1306** may abut, rest on, or couple to the posts **1702**. The top end **1700** is opposite a bottom end **1704** of the battery housing **800**. The batteries of the security camera device **100** may be insertable into the battery housing **800** via the bottom end **1704**. The PCB **808** may include various computing components to enable operation of the security camera device **100**. For PCB **808** may include a Wi-Fi module **1706**, a LORA (long range) module **1708**, and so forth. Shields may be disposed over the Wi-Fi module **1706** and/or the LORA module **1708**. A flex circuit **1710** may also communicatively couple between the port **208** and the PCB **808** for transferring power to components of the security camera device **100**.

(126) FIG. **18** illustrates the battery housing **800**, according to examples of the present disclosure.

The battery housing **800** includes the top end **1700** having the posts **1702** for receiving the PCB **808** and/or the heat dissipating element **1306**, and the bottom end **1704** into which the batteries are insertable.

(127) FIG. **19** illustrates a PCBA assembly **1900** that may include components of the first housing **102** and which resides in the first housing **102**, according to examples of the present disclosure. In some instances, the PCBA assembly **1900** includes the PCBA **810**, the IR LEDS **1108**, the IR LED PCB **1102**, the first LEDs **1200**, the first LED PCB **1104**, the second LEDs **1202**, the second LED PCB **1106**, the camera **802**, the RADAR sensor **806**, the RADAR PCB **1100**, the PIR assembly **1110**, and/or the midframe **1402**. However, the PCBA assembly **1900** may include additional components, such as the microphones **1400**. In some instances, components of the PCBA assembly **1900**, such as the PCBA **810**, couple to the midframe **1402**. In turn, the midframe **1402** may couple to the battery housing **800**. In some instances, the IR LED PCB **1102**, the first LED PCB **1104**, and/or the second LED PCB **1106** are coupled to a first side of the midframe **1402**. The PCBA **810** may couple to an opposite, second side, adjacent to the battery housing **800**.

(128) FIG. **20** illustrates the midframe **1402** of the first housing **102**, or the PCBA assembly **1900**, to which components of the first housing **102** couple, according to examples of the present disclosure. The midframe **1402** may define, or otherwise include various features for receiving components of the first housing **102** or which components of the first housing **102** couple. For example, the midframe **1402** may include the aperture **1404** through which the camera **802** is at least partially disposed and/or an opening **2000** through which the flex circuit **1406** routes.

(129) The midframe **1402** also includes a first receptacle **2002** and a second receptacle **2004** for the first microphone **1400**(1) and the second microphone **1400**(1), respectively. A first flange **2006** and a second flange **2008** of the midframe **1402** also receive the heat dissipating element **1306**, and/or the first LED PCB **1104** and the second LED PCB **1106**, respectively. The first flange **2006** and the second flange **2008** may respectively orient the first LEDs **1200** and the second LEDs **1202** within the environment. For example, in some instances, the midframe **1402** includes a front surface **2010** that is substantially planar (e.g., in the X-Y plane). In some instances, the IR LEDS **1108**, the IR LED PCB **1102**, the RADAR sensor **806**, the RADAR PCB **1100**, the PIR assembly **1110** may couple to the front surface **2010** of the midframe **1402**. The camera **802**, for example, may be disposed behind a rear surface (e.g., opposite the front surface **2010**), of the midframe **1402**.

(130) The first flange **2006** and the second flange **2008** may be oriented at an angle relative to the front surface **2010**. As an example, the first flange **2006** and the second flange **2008** may be angled backward relative to the front surface **2010**. In doing so, the first LEDs **1200** and the second LEDs **1202** are configured to emit light from the first corner and the second corner of the security camera device **100**. In some instances, the first flange **2006** and the second flange **2008** are oriented at between approximately ten degrees and thirty degrees relative to the front surface **2010**.

(131) The midframe **1402** may also include other features for receiving an orienting components of the first housing **102**, and/or to otherwise align the components within the first housing **102**. For example, the midframe **1402** may include prongs **2012** on the first flange **2006** and/or the second flange **2008** for receiving and aligning the heat dissipating element **1306**, the first LED PCB **1104**, and the second LED PCB **1106**, respectively. The midframe **1402** may also include slots **2014** for receiving the PIR assembly **1110**. For example, attachment mechanisms (e.g., hooks, tabs, latches, etc.) may be insertable into the slots **2014** for coupling the PIR assembly **1110** to the midframe **1402**.

(132) FIG. **21** illustrates components of the first housing **102** coupled to the PCBA **810**, according to examples of the present disclosure. For example, the camera **802**, the shields **1504**, the first PIR sensor **804**(1), the second PIR sensor **804**(2), the flex circuit **1406**, and the flex circuit **1502** may couple to or be disposed on the PCBA **810**. As shown, the camera **802** may reside vertically above the PIR sensor(s) **804**. The PCBA **810** may also include holes **2100** that align with tabs, flanges, etc. of the midframe **1402** (e.g., a rear surface of the midframe **1402**) for aligning the PCBA **810** on

the midframe **1402** and/or within the first housing **102**. In some instances, the camera **802** may include a 140 degree horizontal field of view (FoV) and/or an eighty degree vertical FoV. In some instances, the camera **802** may be a 1080P HD camera with high definition resolution (HDR) and color night vision.

(133) FIGS. **22A-22D** illustrate the PIR assembly **1110**, according to examples of the present disclosure. FIG. **22A** illustrates a front perspective view of the PIR assembly **1110**, FIG. **22B** illustrates a front view of the PIR assembly **1110**, FIG. **22C** illustrates a top view of the PIR assembly **1110**, and FIG. **22D** illustrates a cross-sectional view of the PIR assembly **1110** taken along line B-B of FIG. **22B**.

(134) In some instances, the PIR assembly **1110** includes a PIR frame **2200**, a mirror **2202** (e.g., reflective surface), a PIR mount **2214**, and the PIR sensor(s) **804**. As discussed herein, the PIR frame **2200** may include a front **2204** that is oriented towards the front **108** of the security camera device **100**, and a back **2206** (opposite the front **2204**). The PIR sensor(s) **804** and/or the PIR mount **2214** may be disposed at the back **2206** of the PIR frame **2200** (or behind the back **2206** of the PIR frame **2200**). Additionally, the PIR frame **2200** may define a first cavity **2208** and a second cavity **2210**, as well as a divider **2212** disposed between the first cavity **2208** and the second cavity **2210**. The divider **2212** may represent a wall disposed between the first cavity **2208** and the second cavity **2210**.

(135) The first PIR sensor **804(1)** is oriented to detect infrared signals via the first cavity **2208**, and the second PIR sensor **804(2)** is oriented to detect infrared signals via the second cavity **2210**. Collectively, however, the PIR sensor(s) **804** have a field of view (e.g., monitored space) for detecting movements within the environment. For example, if no motion exists in the environment, the PIR sensor(s) **804** may not receive any signals. However, when motion exists, the PIR sensor(s) **804** may receive IR radiation emitted via persons (or animals) within the field of view. In some instances, the first PIR sensor **804(1)** is oriented in a direction away from the bottom **110** of the security camera device **100** (or a bottom of the PIR frame **2200**), and towards the first side **116**. In some instances, the second PIR sensor **804(2)** is oriented in a direction away from the bottom **110** of the security camera device **100** (or a bottom of the PIR frame **2200**), and towards the second side **118**. In some instances, the first PIR sensor **804(1)** and/or the second PIR sensor **804(2)** are oriented away from the divider **2212**.

(136) The mirror **2202** may assist in generating the field of view of the PIR sensor(s) **804**. For example, the PIR sensor(s) **804** may be oriented towards the mirror **2202**, such that the mirror **2202** reflects signals towards the PIR sensor(s) **804**. As shown, the mirror **2202** may be disposed more proximate to a top of the PIR frame **2200** than a bottom of the PIR frame **2200**. Given the compact nature of the PIR assembly **1110** (or more generally, the security camera device **100**), the use of the mirror **2202** may assist in creating a larger field of view for the PIR sensor(s) **804**. In some instances, the mirror **2202** couples to the PIR frame **2200** at a location within the first cavity **2208** and the second cavity **2210**. For example, the mirror **2202** may couple to the PIR frame **2200** at a top of the first cavity **2208** and the second cavity **2210**. As shown, the mirror **2202** may span between the first cavity **2208** and the second cavity **2210**, across the divider **2212**.

(137) The mirror **2202** may include a single mirror that spans between the first cavity **2208** and the second cavity **2210**, across the divider **2212**. However, in some instances, the mirror **2202** may be formed from a first element (e.g., section) and a second element (e.g., section). Here, the first PIR sensor **804(1)** may be oriented towards the first element, and the second PIR sensor **804(2)** may be oriented towards the second element.

(138) The PIR sensor(s) **804** couple to the PIR mount **2214** for disposing or orienting the PIR sensor(s) **804** in a direction towards the front **108** of the security camera device **100**, as well as towards the mirror **2202**. In some instances, the PIR sensor(s) **804** are angled outward from a central vertical plane **2216** (Y-Z plane) of the PIR assembly **1110** (or the PIR frame **2200**). For example, the first PIR sensor **804(1)** may be disposed at an angle α relative to the central vertical

plane **2216**, and/or the second PIR sensor **804(2)** may be disposed at an angle θ relative to the central vertical plane **2216**. In some instances, the angle α and/or the angle θ may be between ten degrees and forty-five degrees. Disposing the first PIR sensor **804(1)** and the second PIR sensor **804(2)** at the angle α and the angle θ , respectively, may increase a field of view of the PIR sensor(s) **804** for detecting motion. More particularly, disposing the first PIR sensor **804(1)** and the second PIR sensor **804(2)** at the angle α and the angle θ , respectively, may increase a horizontal component of the field of view. In some instances, the angle α and the angle θ may be the same and/or different. Moreover, as discussed herein, the PIR mount **2214** may orient the first PIR sensor **804(1)** and the second PIR sensor **804(2)** at the angle α and the angle θ , respectively.

(139) In addition to being oriented outward, the PIR sensor(s) **804** may be oriented vertically upward, relative to a central horizontal plane **2218** of the PIR assembly **1110** (or the PIR frame **2200**). The central horizontal plane **2218** may be disposed centrally through the PIR frame **2200**, and/or may be orthogonal to the central vertical plane **2216**. As shown in FIG. **22D**, the second PIR sensor **804(2)** may be oriented at an angle β relative to the central horizontal plane **2218**, which extends centrally (e.g., in the Y-direction) through the PIR frame **2200**. In some instances, the angle β may be between approximately ten degrees and thirty degrees. Disposing the second PIR sensor **804(2)** at the angle β orients the second PIR sensor **804(2)** towards the mirror **2202**, such that the second PIR sensor **804(2)** may receive signals (e.g., IR radiation) reflected off the mirror **2202**. In some instances, the second PIR sensor **804(2)** may be oriented substantially parallel with the mirror **2202**. Alternatively, in some instances, the second PIR sensor **804(2)** may not be oriented parallel with the mirror **2202**. For example, the mirror **2202** may be oriented at an angle relative to the central horizontal plane **2218**, which may be between ten degrees and thirty degrees. Although the discussion is with regard to the second PIR sensor **804(2)**, the first PIR sensor **804(1)** may be oriented at the angle β as well.

(140) In some instances, the PIR assembly **1110** couples to the PCBA **810**. Additionally, the second lens **130** may be disposed over the front **2204** of the PIR assembly **1110**, where the PIR sensor(s) **804** are arranged towards the second lens **130**. In some instances, an IR-aperture height of the PIR sensor(s) **804** may be twelve millimeters (mm) (e.g., in the Y-direction), and an IR-aperture width of the PIR sensor(s) may be thirty-eight mm. In this sense, the PIR sensor(s) **804** are arranged to receive signals through the IR-aperture for detecting motion within the environment. However, other sizes of the IR-aperture, whether the width and/or the height, may be different. In some instances, a portion of the first PIR sensor **804(1)** may be disposed within the first cavity **2208**, and/or a portion of the second PIR sensor **804(2)** may be disposed in the second cavity **2210**.

(141) FIGS. **23A-23D** illustrate the PIR frame **2200**, according to examples of the present disclosure. FIG. **23A** illustrates a front perspective view of the PIR frame **2200**, FIG. **23B** illustrates a rear perspective view of the PIR frame **2200**, FIG. **23C** illustrates a front view of the PIR frame **2200**, and FIG. **23D** illustrates a side view of the PIR frame **2200**.

(142) The PIR frame **2200** includes a top **2300**, a bottom **2302**, a first side **2304**, a second side **2306**, a front **2308**, and a back **2310**. When disposed in the first housing **102**, the top **2300** may be oriented towards the top **2300** of the security camera device **100**, the bottom **2302** may be oriented towards the bottom **114** of the security camera device **100**, the front **2308** may be oriented towards the front **108** of the security camera device **100**, and the back **2310** may be oriented towards the PCBA **810** (or towards the back **108** of the security camera device **100**). In some instances, the front **2308** of the PIR frame **2200** may correspond to the front **2204** of the PIR assembly **1110**. Additionally, the first side **2304** and the second side **2306** may be oriented towards the lateral sides of the security camera device **100**. As shown, the PIR frame **2200** may generally include a rectangular shape.

(143) As introduced above, the PIR frame **2200** defines the first cavity **2208** and the second cavity **2210** in which the first PIR sensor **804(1)** and the second PIR sensor **804(2)** are at least partially disposed, as well as the divider **2212** between the first cavity **2208** and the second cavity **2210**.

However, in some instances, the PIR sensor(s) **804** may not be disposed within the first cavity **2208** and the second cavity **2210**, respectively, but may be oriented towards the first cavity **2208** and the second cavity **2210**, respectively (e.g., spaced apart from the back **2310**). For example, the PIR sensor(s) **804** may be disposed behind the back of the PIR frame **2200**, and oriented towards the first cavity **2208** and the second cavity **2210**, respectively. In some instances, the divider **2212** is centrally located between the first side **2304** and the second side **2306**. The first cavity **2208** and the second cavity **2210** may be the same or of different sizes.

(144) The PIR frame **2200** may also include an exterior surface **2312** and an interior surface **2314**. In some instances, the first cavity **2208**, the second cavity **2210**, and the divider **2212** are formed within or by the interior surface **2314**. When the mirror **2202** couples to the PIR frame **2200**, the mirror **2202** may be disposed along a top of the first cavity **2208** and the second cavity **2210**. In some instances, the mirror **2202** may couple to the top via adhesives, fasteners, snap-fits, and so forth. The mirror **2202** may also span between the first cavity **2208** and the second cavity **2210**, across the divider **2212**. In some instances, the interior surface **2314** along a top of the first cavity **2208** and a top of the second cavity **2210** may include an angle (relative to the central horizontal plane **2218**) such that the mirror **2202** is oriented as described herein.

(145) In some instances, the exterior surface **2312** forms alignment mechanisms and/or attachment mechanisms for coupling the PIR frame **2200** (or the PIR assembly **1110**) to the PCBA **810** (or more generally, the first housing **102**). For example, the back **2310** may include hooks **2316** that engage with slots or grooves in the PCBA **810** for coupling the PIR frame **2200** to the PCBA **810**. A groove **2318** may also be disposed on the exterior surface **2312** for receiving a gasket or other seal (not shown). As shown, the back **2310** may also include openings for receiving the PIR sensor(s) **804** and through which the PIR sensor(s) **804** receive signals.

(146) Additional details of the PIR assembly **1110**, or PIR sensor(s) configured to detect motion, are described in, for example, U.S. patent application Ser. No. 17/855,752, filed Jun. 30, 2022, the entirety of which is herein incorporated by reference.

(147) FIGS. **24A** and **24B** illustrate the PIR sensor(s) **804** coupled to the PIR mount **2214**, according to examples of the present disclosure. FIG. **24A** illustrates a top view with the PIR sensor(s) **804** coupled to the PIR mount **2214**, while FIG. **24B** illustrates a side view with the PIR sensor(s) **804** coupled to the PIR mount **2214**.

(148) The first PIR sensor **804(1)** is shown including first prongs **2400** that are disposed through the PIR mount **2214** for communicatively coupling to the PCBA **810**, and the second PIR sensor **804(2)** is shown including second prongs **2402** that are disposed through the PIR mount **2214** for communicatively coupling to the PCBA **810**. As discussed herein, the PIR mount **2214** may orient the first PIR sensor **804(1)** at the angle α and the second PIR sensor **804(2)** at the angle θ . The PIR mount **2214** may also orient the first PIR sensor **804(1)** and/or the second PIR sensor **804(2)** at the angle β .

(149) FIGS. **25A** and **25B** illustrate the PIR mount **2214**, according to examples of the present disclosure. FIG. **25A** illustrates a front perspective view with the PIR mount **2214**, while FIG. **25B** illustrates a side view of the PIR mount **2214**.

(150) The PIR mount **2214** may include a front **2500** having a first portion **2502** and a second portion **2504**. The first portion **2502** may receive the first PIR sensor **804(1)**, such as a back of the first PIR sensor **804(1)**, while the second portion **2504** may receive the second PIR sensor **804(2)**, such as a back of the second PIR sensor **804(2)**. The first portion **2502** is shown including first channels **2506** through which the first prongs **2400** may be at least partially disposed through. The second portion **2504** is shown including second channels **2508** through which the second prongs **2402** may be at least partially disposed through.

(151) In some instances, the first portion **2502** and/or the second portion **2504** may be recessed from the front **2500** (e.g., in the Z-direction) within an exterior surface such that the first PIR sensor **804(1)** and the second PIR sensor **804(2)** may be at least partially encapsulated by the PIR

mount **2214**. For example, the first portion **2502** and/or the second portion **2504** may resemble pockets that receive the PIR sensor(s) **804**, respectively. As shown, the first portion **2502** and the second portion **2504** (or more generally the PIR mount **2214**) may be configured to orient the first PIR sensor **804(1)** and the second PIR sensor **804(2)** as described herein. The PIR mount **2214** may also include hooks **2510** that couple the PIR mount **2214** to the PIR frame **2200** or the PCBA assembly **1900**.

(152) FIGS. **26A** and **26B** illustrate the second lens **130** that is configured to be disposed over the PIR sensor(s) **804** of the security camera device **100**, according to examples of the present disclosure. FIG. **26A** illustrates a front perspective view of the second lens **130**, and FIG. **26B** illustrates a cross-sectional view of the second lens **130**, taken through line C-C of FIG. **26A**.

(153) The second lens **130** may include channels **2600** for engaging with features disposed on the exterior surface **2312** of the PIR frame **2200**. In some instances, the PIR frame **2200** is configured to reside at least partially within a cavity **2602** of the second lens **130**. For example, the PIR frame **2200** (and components coupled to the PIR frame **2200**), may be disposed through a rear **2604** of the second lens **130** so as to be disposed within the cavity **2602**.

(154) In some instances, the second lens **130** may include different ranks (e.g., focal length, aperture, etc.). For example a first portion **2606** of the second lens **130** may include a first rank, while a second portion **2608** of the second lens **130** may include a second rank. In some instances, the first rank and the second rank may be different. Additionally, in some instances, the first portion **2606** may represent a greater portion of the second lens **130**, as compared to the second portion **2608** of the second lens. For example, the first portion **2606** may be eleven mm in height, while the second portion **2608** may be three mm in height (Y-direction).

(155) FIG. **27** illustrates a partially exploded view of the first housing **102**, showing the cap **122** being disposed above the first cover **120**. The cap **122** may be removed from a top of the first housing **102** to expose an interior **2700** of the first housing **102**. Within the interior **2700**, components of the security camera device **100** reside. In some instances, the cap **122** and the first housing **102** may couple together via complementary latching mechanisms (e.g., keyways, slots, tabs, hooks, etc.). For example, the cap **122** may include latching mechanisms **2702** disposed around a perimeter (e.g., periphery) of the cap **122** that engage with second latching mechanisms disposed around a perimeter of an interior surface **2704** of the first cover **120**.

(156) Removing the cap **122** exposes the PCB **808**, including the Wi-Fi module **1706** and the LORA module **1708**. However, various other components may be mounted to, or communicatively coupled to the PCB **808**. The flex circuit **1406** is also shown extending from the PCB **808** for routing to and coupling with the PCBA **810**. The flex circuit **1502** extends from the PCB **808** for routing to and coupling with the port **208**. Additionally, the first wire **1308** and the second wire **1310** communicatively couple the PCB **808** with the first LED PCB **1104** and the second LED PCB **1106**, respectively.

(157) The cap **122** is also shown including the button **136**, which may engage with components of the PCB **808** such that presses (or actuation) of the button **136** are sensed.

(158) FIGS. **28A** and **28B** illustrate the PCB **808**, according to examples of the present disclosure. The PCB **808** may include a first side **2800** and a second side **2802**. The first side **2800** may be oriented towards the cap **122**, or the top **112** of the security camera device **100**, while the second side **2802** may be oriented towards the top end **1700** of the battery housing **800** or the bottom **114** of the security camera device **100**.

(159) The PCB **808** includes the Wi-Fi module **1706** and the LORA module **1708**. The Wi-Fi module **1706** may communicatively couple to a Wi-Fi antenna **2804** disposed on the first side **2800** of the PCB **808**. The Wi-Fi antenna **2804** may extend from the first side **2800** of the PCB **808** and be oriented towards the interior surface **2704** of the first cover **120** when the security camera device **100** is assembled. The LORA module **1708** may communicatively couple to the antenna **1500**. The first side **2800** may also include various connectors for receiving the flex circuit **1502**, the flex

circuit **2710**, the first wire **1308**, and the second wire **1310**.

(160) The second side **2802** of the PCB **808** may include a Wi-Fi driver **2806**, which in some instances, is communicatively coupled to the Wi-Fi module **1706** disposed on the first side **2800**. A DC-DC converter **2808** may also be disposed on the second side **2802**. The second side **2802** may additionally include a shielding plate **2810** disposed over component circuitry (e.g., processor(s), memory, capacitors, resistors, etc.). In some instances, the shielding plate **2810** is shown being located between the Wi-Fi driver **2806** and the DC-DC converter **2808**.

(161) The PCB **808** includes first prongs **2812** and second prongs **2814** corresponding to charging contacts of the security camera device **100**. The first prongs **2812** and the second prongs **2814** may engage with contacts (e.g., pads) on the top end **1700** of the battery housing **800** for transferring power from the batteries to the PCB **808** (and to additional computing components of the security camera device **100**). In some instances, the first prongs **2812** receive power from a first battery within the battery housing **800**, while the second prongs **2814** receive power from a second battery within the battery housing **800**.

(162) The PCB **808** may also include alignment orifices **2816** that align the PCB **808** with the battery housing **800**. For example, the posts **1702** may extend at least partially through the alignment orifices **2816** for aligning the PCB **808** on the battery housing **800**. The alignment orifices **2816** may extend through the PCB **808**, between the first side **2800** and the second side **2802**.

(163) FIG. **29** illustrates the interior **2700** of the first housing **102**, showing components of the first housing **102** being removed, according to examples of the present disclosure. The interior surface **2704** of the first cover **120** includes various gussets, tabs, flanges, and so forth for receiving components that reside within the interior surface **2704**. For example, the various components that reside within the first housing **102** may couple to, rest on, or otherwise engage with the gussets, tabs, flanges, and so forth for aligning the components within the first housing **102**.

(164) FIGS. **30A** and **30B** illustrate bottom views of the first housing **102**, according to examples of the present disclosure. The first cover **120** forms the first attachment mechanisms **900** that engage with the second attachment mechanisms **902** of the frame **904** of the second housing **104**. The first attachment mechanisms **900** are shown being disposed around the perimeter **908** of the first cover **120**.

(165) The security camera device **100** includes batter(ies) **3000**, such as a first battery **3000(1)** and a second battery **3000(2)**, for power components of the security camera device **100**. The batter(ies) **3000** are insertable into the battery housing **800** via the bottom end **1704**. The battery housing **800** may define a first battery receptacle **3002** and a second battery receptacle **3004** for receiving the first battery **3000(1)** and the second battery **3000(2)**, respectively. A first tab **3006** and a second tab **3008** may engage with the first battery **3000(1)** and the second battery **3000(2)** (e.g., ends thereof), respectively. For example, the first tab **3006** and the second tab **3008** may engage with the first battery **3000(1)** and the second battery **3000(2)** to secure the first battery **3000(1)** and the second battery **3000(2)** within the first receptacle **3002** and the second receptacle **3004**, respectively. Pressing the first tab **3006** and the second tab **3008** (e.g., in the X-direction and towards a center of the security camera device **100**) may permit the first battery **3000(1)** and the second battery **3000(2)** to be pulled out of the first battery receptacle **3002** and the second battery receptacle **3004**, respectively. Although two battery receptacles are shown, the battery housing **800** may include more than or less than two battery receptacles. Additionally, the security camera device **100** may not include two batteries, but in some instances, may operate via a single battery. In some instances, the batter(ies) **3000** are arranged more proximate to the back **110** of the security camera device **100** than the front **108** given the placement of the PCBA assembly **1900** in front of the batter(ies) **3000**.

(166) The first housing **102** includes a first contact **3010** and a second contact **3012** for engaging with corresponding contacts of the second housing **104**. When the first housing **102** and the second

housing **104** are coupled together, the first contact **3010** and the second contact **3012** engage with corresponding contacts of the second housing **104** for communicatively coupling computing components in the first housing **102** with computing components in the second housing **104**. Such communicative coupling may transfer power, signals, data, and so forth. In some instances, the first contact **3010** and the second contact **3012** are coupled to an amplifier within the first housing **102** that drives the loudspeaker **814** in the second housing. As shown, the first contact **3010** and the second contact **3012** may be arranged proximate to the front **108** of the security camera device **100**, however, the first contact **3010** and the second contact **3012** may be located elsewhere on the first housing **102**. The use of the contacts to communicatively couple the first housing **102** and the second housing **104** may eliminate the need for wires extending between the first housing **102** and the second housing **104**. This may enable the first housing **102** and the second housing **104** to be conveniently decoupled, for example, during a replacement of the batter(ies) **3000**.

(167) As shown, the first contact **3010** and the second contact **3012** may represent circuit traces having a length for receiving contacts of the second housing **104**. In some instances, the first contact **3010** and the second contact **3012** include a curvature given the rotatable nature for coupling the first housing **102** and the second housing **104**. For example, during a twisting of the second housing **104** the contacts of the second housing **104** may translate along the first contact **3010** and the second contact **3012**. In some instances, the first contact **3010** and/or the second contact **3012** are disposed on or within the bottom end **1704** of the battery housing **800**. The first contact **3010** and the second contact **3012**, although not shown, may communicatively couple to the PCB **808** and/or the PCBA **810** (as well as other components of the first housing **102**). In some instances, the first contact **3010** and the second contact **3012** include the same length, size, and/or shape.

(168) FIGS. **31A-31C** illustrate the first cover **120**, according to examples of the present disclosure. FIG. **31A** illustrates a front planar view of the first cover **120**, FIG. **31B** illustrates a first side planar view of the first cover **120**, and FIG. **31C** illustrates a second side planar view of the first cover **120**.

(169) The first cover **120** defines the first attachment mechanisms **900** arranged at a bottom end **3100** of the first cover **120**. A top end **3102**, opposite the bottom end **3100**, is open to receive the cap **122**. In some instances, the first cover **120** is formed via a unitary body, or may be multiple bodies (e.g., halves) coupled together.

(170) The first cover **120** also defines various receptacles for receiving the lenses of the first housing **102**. For example, a first receptacle **3104** may receive the first lens **128**, a second receptacle **3106** may receive the second lens **130**, a third receptacle **3108** may receive the third lens **132**, and a fourth receptacle **3110** may receive the fourth lens **134**. The receptacles are sized and configured to receive the lenses, respectively. The first cover **120** may also define various keyways, slots, and so forth into which or within which the lenses engage. For example, the lenses may couple to the first cover **120** via press and fit mechanisms, adhesives, and so forth. Such keyways and slots may also align the lenses on the first cover **120**. The first receptacle **3104** and the second receptacle **3106** may be located on a front of the first cover **120**, while the third receptacle **3108** and the fourth receptacle **3110** may be located on the front and/or sides of the first cover **120**.

(171) The receptacles may represent cutouts within an exterior surface **3112** of the first cover **120** such that when the lenses couple to the first cover **120**, a smooth or continuous surface is formed. For example, the receptacles may include a depth that substantially corresponds to a thickness of the lenses. Once the lenses couple to the first cover **120**, the lenses may align with the exterior surface **3112**.

(172) FIGS. **32A-32C** illustrate various views of the second housing **104**, according to examples of the present disclosure. FIG. **32A** illustrates a front perspective view of the second housing **104**, FIG. **32B** illustrates a rear perspective view of the second housing **104**, and FIG. **32C** illustrates a top planar view of the second housing **104**.

(173) As introduced above, the second housing **104** includes the frame **904**, the second cover **124**, and the loudspeaker grill **138**. In some instances, the second cover **124** couples to the frame **904** via fasteners or other mechanisms (e.g., press-fit), and/or the loudspeaker grill **138** couples to the second cover **124** via press-fit mechanisms. The second attachment mechanisms **902** may be defined by or within the annulus **906** of the frame **904**.

(174) The second housing **104** includes contacts for engaging with the first contact **3010** and the second contact **3012** of the first housing **102**. For example, the second housing **104** may include a first contact **3200** and a second contact **3202** that engage with the first contact **3010** and the second contact **3012**, respectively. In some instances, the first contact **3010** and the second contact **3012** may represent prongs, pins, and the like that engage with the traces of the first contact **3010** and the second contact **3012**, respectively. However, although the first contact **3010** and the second contact **3012** are described as being traces, and the first contact **3200** and the second contact **3202** are described as being prongs, in some instances, the first contact **3010** and the second contact **3012** may represent prongs and the first contact **3200** and the second contact **3202** may represent traces. In accordance with one or more preferred implementations, the first contact **3200** and the second contact **3202** may have a shape and be constructed from a material allowing the first contact **3200** and the second contact **3202** to act as a spring, as illustrated for example in FIGS. 33B and 33C (in which some elements are omitted from the illustrations for clarity).

(175) The first contact **3200** and the second contact **3202** may be biased to engage with the first contact **3010** and the second contact **3012** (e.g., in the Y-direction). For example, springs may bias the first contact **3200** and the second contact **3202** in a direction towards the first contact **3010** and the second contact **3012** in order to engage with the first contact **3010** and the second contact **3012**, respectively. The first contact **3200** and the second contact **3202** may be disposed through a portion of the frame **904**. For example, when the bottom end **3100** of the first cover **120** (or a bottom of the first housing **102**) is inserted into the annulus **906** of the frame **904**, the first contact **3010** and the second contact **3012** may engage with the first contact **3200** and the second contact **3202**, respectively. In some instances, the first contact **3010** and the second contact **3012** may engage with the first contact **3200** and the second contact **3202** during rotation of the first housing **102** and/or the second housing **104**.

(176) The second housing **104** may include alignment tabs **3204** that engage with features of the first housing **102**. The alignment tabs **3204** may serve to align the first housing **102** and the second housing **104** such that the first attachment mechanisms **900** and the second attachment mechanisms **902** may engage with one another.

(177) FIGS. 33A-33C illustrates the loudspeaker **814** of the security camera device **100**, according to examples of the present disclosure. The loudspeaker **814** couples to the frame **904** of the second housing **104** for orienting the loudspeaker **814** to fire in a direction towards the front **108** of the security camera device **100**. As shown, the loudspeaker **814** may couple to a bottom of the frame **904**, opposite a top that receives the first housing **102**. Terminal ends **3300** of the first contact **3200** and the second contact **3202** are shown being disposed through the frame **904**. Wires, connectors, and so forth may extend from the terminal ends **3300** to the loudspeaker **814**, as well as other components with the second housing **104**.

(178) In some instances, portions of the first contact **3200** and the second contact **3202** are disposed within a housing **3202**, between the terminal ends **3300** and within the frame **904**. Wires may route between the terminal ends **3200** and the loudspeaker **814**.

(179) FIGS. 34A-34D illustrate the frame **904** of the second housing **104**, according to examples of the present disclosure. FIG. 34A may illustrate a front perspective view of the frame **904**, FIG. 34B may illustrate a bottom planar view of the frame **904**, FIG. 34C may illustrate a top planar view of the frame **904**, and FIG. 34D may illustrate a side view of the frame **904**.

(180) The frame **904** may include a top **3400** and a bottom **3402** opposite the top **3400**. When the security camera device **100** is assembled, the top **3400** may be oriented towards the first housing

102 (or the top **112** of the security camera device **100**), while the bottom **3402** may be oriented towards the bottom **114** of the security camera device **100**. The top **3400** may define a receptacle **3414** into which at least a portion of the first housing **102** is received. The receptacle **3414** also includes the annulus **906** that has the second attachment mechanisms **902**. The second cover **124** is configured to reside over at least a portion of an exterior surface **3404** of the frame **904**. For example, the second cover **124** may couple to sides of the frame **904**. The sides of the frame **904** may include various tabs, keyways, features, and so forth for receiving and/or otherwise aligning the second cover **124** onto the frame **904**.

(181) A channel **3406** extends through the frame **904**, between the top **3400** and the bottom **3402**. The first contact **3200** and the second contact **3202** (e.g., the terminal ends **33000**) may be at least partially disposed within the channel **3406**, and route power, signals, and so forth to and from the loudspeaker **814**, for example. The bottom **3402** may also include a mount **3408** for receiving the loudspeaker **814**. As shown in FIG. **34D**, the mount **3408** may include a mounting surface **3410** for disposing the loudspeaker **814** at the orientation described herein. That is, when an end of the loudspeaker **814** couples to the mounting surface **3410**, the loudspeaker **814** may be oriented such that the loudspeaker **814** emits sounds outward and away from the security camera device **100** (e.g., in front of and to the sides of the security camera device **100**). The frame **904** may further define the alignment tabs **3204**.

(182) FIGS. **35A-35C** illustrate the second cover **124** of the second housing **104**, according to examples of the present disclosure. FIG. **35A** may illustrate a front perspective view of the second cover **124**, FIG. **35B** may illustrate a front planar view of the second cover **124**, and FIG. **35C** may illustrate a side view of the second cover **124**.

(183) The second cover **124** is configured to couple to the frame **904**, over the loudspeaker **814**. For example, the second cover **124** may be disposed at least partially over the exterior surface **3404** of the frame **904**. When coupled to the frame **904**, the loudspeaker **814** is configured to reside within a cavity of the second cover **124**. The second cover **124** is shown including orifices **3500** that permit sound generated by the loudspeaker **814** to exit the interior of the security camera device **100**. The second cover **124** also includes various features for receiving the loudspeaker grill **138**. The second cover **124** includes an exterior surface **3502** that corresponds to an exterior surface of the security camera device **100**.

(184) The second housing **104** is shaped to accommodate the orientation of the loudspeaker **814**. For example, a front of the second cover **124**, which may correspond to the front **108** of the security camera device **100**, may include a first height **3504**, while a back of the second cover **124**, which may correspond to the back **110** of the security camera device **100**, may include a second height **3506**. The second height **3506** may be greater than the first height **3504** such that the loudspeaker **814** is permitted to mount to the mounting surface **3410** and oriented accordingly.

(185) FIG. **36** illustrates a perspective view of the loudspeaker grill **138**, according to examples of the present disclosure. The loudspeaker grill **138** is configured to couple to the frame **904** or at the bottom of the security camera device **100**. The loudspeaker grill **138** may substantially prevent the permeation of liquid, debris, and so forth into the security camera device **100**, but may permit sound to travel outward from the security camera device **100**. The loudspeaker grill **138** includes an interior surface **3600** that resides over a portion of the second cover **124**. When coupled to the second cover **124**, an exterior surface **3602** of the loudspeaker grill **138** may align (e.g., flush) with the exterior surface **3502** of the second cover **124**. The loudspeaker grill **138** is further shown including attachment mechanisms **3604** for coupling the loudspeaker grill **138** the frame **904** and/or the second housing **104**.

(186) In some instances, the loudspeaker grill **138** may conceal the loudspeaker **814**, as well as other components of the second housing **104**, such as the orifices **3500**, such that no internal structure or components can be seen through the loudspeaker grill **138**.

(187) In accordance with one or more preferred implementations, the first contact **3010** and the

second contact **3012** of the first housing **102** couple with the first contact **3200** and the second contact **3202** of the second housing **104** to provide a signal over the contacts to the loudspeaker **814**. In accordance with one or more preferred implementations, an amplifier is disposed within the first housing **102**, and the signal provided to the loudspeaker **814** is an amplified signal generated using the amplifier that controls audio output of the loudspeaker **814**.

(188) In accordance with one or more preferred implementations, the first housing **102** and the second housing **104** may include more or less contacts, and power may be provided to components of the second housing **102** via coupling of one or more contacts of the first housing **102** to one or more contacts of the second housing **104**. In accordance with one or more preferred implementations, data or an audio signal may be provided to components of the second housing **104** via coupling of one or more contacts of the first housing **102** to one or more contacts of the second housing **104**. In accordance with one or more preferred implementations, an amplifier is disposed within the second housing **104**.

(189) FIG. **37** illustrates select components of the security camera device **100**, according to examples of the present disclosure. The security camera device **100** is shown including processor(s) **3700** and memory **3702**, where the processor(s) **3700** may perform various functions associated with controlling an operation of the security camera device **100**, and the memory **3702** may store instructions executable by the processor(s) **3700** to perform the operations described herein.

(190) The security camera device **100** includes camera(s) **802** for capturing image/video data **3704** within an environment of the security camera device. In some instances, the camera(s) **802** may include red, green, blue, depth (RGBD) camera(s) and/or three-dimensional (3D) sensors. Additionally, the security camera device **100** may include any other sensor(s) **3706** (e.g., ambient light sensor) that generates sensor data **3708**. Further, the security camera device **100** may include the PIR sensor(s) **804** and the RADAR sensor **806** that generates the sensor data **3708**. In some instances, the PIR sensor(s) **804** act as motion sensors for detecting movement within a field of view. The PIR sensor(s) **804** may be secured on or within a PIR frame that may reside behind a lens (e.g., a Fresnel lens) of the security camera device **100**. In such examples, the PIR sensor(s) **804** may detect IR radiation in a field of view of the PIR sensor(s) **804**, and produce an output signal (e.g., voltage) that changes as the amount of IR radiation in the field of view changes. The amount of voltage in the output signal may be compared, by the processor(s) **3700**, for example, to one or more threshold voltage values to determine if the amount of voltage in the output signal is indicative of motion, and/or if the amount of voltage in the output signal is indicative of motion of an entity that is to be recorded by the camera(s) **802**. In some instances, the PIR sensor(s) **804** may detect the motion for activating the camera(s) **802** and/or a microphone(s) **1400** to begin capturing image data and/or audio data, respectively.

(191) In some instances, the PIR sensor(s) **804** is/are used to detection motion within an environment of the security camera device **100**. However, in some instances, the camera(s) **802** and/or the RADAR sensor **806**, in addition to or alternative from the PIR sensor(s) **804**, may be used to detect motion. For example, computer vision techniques may be used to detect objects of interest. In some instances, the camera(s) **802** may include a CMOS image sensor, and a digital processor that may perform embedded processing within the low-power CVM itself, such that the low-power CVM may output post-processed computer vision metadata to the processor(s) **3700**. The metadata may include information such as the presence of a particular type of entity (e.g., person, animal, vehicle, parcel, etc.), a direction of movement of the entity, a distance of the entity from the security camera device **100**, etc. As a result of including the computer vision, the security camera device **100** may leverage computer vision to implement computer vision for one or more aspects, such as motion detection, object recognition, and/or facial recognition. Computer vision includes methods for acquiring, processing, analyzing, and understanding images and, in general, high-dimensional data from the real world in order to produce numerical or symbolic information, e.g., in the form of decisions. Computer vision seeks to duplicate the abilities of human vision by

electronically perceiving and understanding an image.

(192) The security camera device **100** also includes LED(s) **3710**, such as IR LEDs and/or white LEDs, for illuminating and/or emitting light within the environment of the security camera device **100**. Any number of IR LEDs and/or white LEDs may be included, and the IR LEDs and the white LEDs may be arranged about various sides of the security camera device **100** (e.g., front, sides, etc.). In some instances, in response to the PIR sensor(s) **804**, the camera(s) **802**, and/or the RADAR sensor **806** detecting motion, the LED(s) **3710** may receive an output signal from the processor(s) **3700** that causes the LED(s) **3710** to activate the one or more lights. The IR LEDs may also be used to detect motion and/or record image/video data **3704** in low-light conditions.

(193) The security camera device **100** includes the microphone(s) **1400** that generate audio data **3712**. The microphone(s) **1400** may include an array of microphone for beamforming audio signals within the environment of the security camera device **100**. Loudspeaker(s) **814** may output sound in a direction away from the security camera device **100**. The sound output by the Loudspeaker(s) **814** may include the audio data **3712**, which may be received from one or more communicatively coupled devices, or other audio (e.g., siren, alarm, etc.).

(194) Network interface(s) **3714** permit the security camera device **100** to communicate over one or more networks. Example network interface(s) **3714** include, without limitation, Wi-Fi, Bluetooth, ZigBee, Bluetooth Low Energy (BLE), LTE, and so forth. The network interface(s) **3714** permit communication with remote device(s), such as mobile devices (e.g., phone), systems (e.g., cloud), and so forth. The network(s) may be representative of any type of communication network, including data and/or voice network, and may be implemented using wired infrastructure (e.g., cable, CAT5, fiber optic cable, etc.), a wireless infrastructure (e.g., RF, cellular, microwave, satellite, Bluetooth, etc.), and/or other connection technologies.

(195) In some instances, inbound data from may be routed through the network interface(s) **3714** before being directed to the processor(s) **3700**, and outbound data from the processor(s) **3700** may be routed through the network interface(s) **3714**. The network interface(s) **3714** may therefore receive inputs, such as data, from the processor(s) **3700**, the camera(s) **802**, the PIR sensor(s) **804**, the RADAR sensor **806**, the button(s) **136**, and so forth. For example, when button(s) **136** receive touch input, the network interface(s) **3714** may perform one or more functions, such as to transmit a signal over a wireless connection and/or a wired connection. As another example, the network interface(s) **3714** may be configured to transmit data to and/or receive data from one or more network devices. The network interface(s) **3714** may act as a conduit for data communicated between various components and the processor(s) **3700**.

(196) The security camera device **100** may include the button(s) **136** for controlling an operation of the security camera device **100**, such as power button, a wireless connectivity button, a mute button, volume buttons, sync buttons, or any other type of button or control. The button(s) **136**, may in some instances, be multi-function buttons. The button(s) **136** may be mechanical (e.g., having physically movable components) and/or electronic (e.g., capacitive sensors, optical sensors, touch screen, or the like).

(197) Batter(ies) **3000** may power the components of the security camera device **100**. Any number of batter(ies) **3000** may be included, and the batter(ies) **3000** may be rechargeable/replaceable. Additionally, the security camera device **100** may be powered using a source of external AC (alternating-current) power, such as mains power. When the batter(ies) **3000** are depleted, the batter(ies) **3000** may be recharged by connecting a power source to the batter(ies) **3000** (e.g., using a USB connector).

(198) Although certain components of the security camera device **100** are illustrated, it is to be understood that the security camera device **100** may include additional or alternative components. For example, the security camera device **100** may include other input/output devices (e.g., display screen), heat dissipating elements, computing components (e.g., PCBs), antennas, ports (e.g., USB), and so forth).

(199) As used herein, a processor, such as the processor(s) **3700** may include multiple processors and/or a processor having multiple cores. Further, the processor(s) may comprise one or more cores of different types. For example, the processor(s) may include application processor units, graphic processing units, and so forth. In one implementation, the processor(s) may comprise a microcontroller and/or a microprocessor. The processor(s) may include a graphics processing unit (GPU), a microprocessor, a digital signal processor or other processing units or components known in the art. Alternatively, or in addition, the functionally described herein can be performed, at least in part, by one or more hardware logic components. For example, and without limitation, illustrative types of hardware logic components that may be used include field-programmable gate arrays (FPGAs), application-specific integrated circuits (ASICs), application-specific standard products (ASSPs), system-on-a-chip systems (SOCs), complex programmable logic devices (CPLDs), etc. Additionally, each of the processor(s) may possess its own local memory, which also may store program components, program data, and/or one or more operating systems.

(200) Memory, such as the memory **3702** may include volatile and nonvolatile memory, removable and non-removable media implemented in any method or technology for storage of information, such as computer-readable instructions, data structures, program component, or other data. Such memory may include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, RAID storage systems, or any other medium that which can be used to store the desired information and which can be accessed by a computing device. The memory may be implemented as computer-readable storage media (“CRSM”), which may be any available physical media accessible by the processor(s) to execute instructions stored on the memory. In one basic implementation, CRSM may include random access memory (“RAM”) and Flash memory. In other implementations, CRSM may include, but is not limited to, read-only memory (“ROM”), electrically erasable programmable read-only memory (“EEPROM”), or any other tangible medium that can be used to store the desired information and which can be accessed by the processor(s).

(201) While the foregoing invention is described with respect to the specific examples, it is to be understood that the scope of the invention is not limited to these specific examples. Since other modifications and changes varied to fit particular operating requirements and environments will be apparent to those skilled in the art, the invention is not considered limited to the example chosen for purposes of disclosure, and covers all changes and modifications which do not constitute departures from the true spirit and scope of this invention.

(202) Although the application describes embodiments having specific structural features and/or methodological acts, it is to be understood that the claims are not necessarily limited to the specific features or acts described. Rather, the specific features and acts are merely illustrative some embodiments that fall within the scope of the claims of the application.

Claims

1. An electronic device, comprising: a first housing having a first end and a second end, the first housing including: a microphone disposed proximate the first end of the first housing, a camera disposed further from the first end of the first housing than the microphone, a radar sensor disposed further from the first end of the first housing than the microphone, a first frame having a first side and a second side opposite the first side, a printed circuit board assembly (PCBA) disposed on the first side of the first frame, a passive infrared (PIR) assembly disposed further from the first end of the first housing than then radar sensor, the PIR assembly having: a second frame disposed on the second side of the first frame, the second frame defining at least a first cavity and a second cavity, a mount coupled to the PCBA, a mirror coupled to the second frame at a location within the first cavity and the second cavity, a first PIR sensor coupled to the mount and oriented towards the first

cavity, and a second PIR sensor coupled to the mount and oriented towards the second cavity; a second housing coupled to the second end of the first housing, the second housing including a speaker; and a mount coupled to the first housing for disposing the electronic device on a surface.

2. The electronic device of claim 1, wherein: the second frame includes a central horizontal plane and a central vertical plane; the first PIR sensor is oriented in a first direction that is angled away from the central horizontal plane and the central vertical plane; and the second PIR sensor is oriented in a second direction that is angled away from the central horizontal plane and the central vertical plane.

3. The electronic device of claim 1, wherein: the second frame includes a wall disposed between the first cavity and the second the first cavity includes a first top and a first bottom opposite the first top; the second cavity includes a second top and a second bottom opposite the second the location within the first cavity and the second cavity that the mirror couples to includes the first top and the second top; the first PIR sensor is oriented in a first direction angled away from the first bottom; and the second PIR sensor is oriented in a second direction angled away from the second bottom.

4. A device, comprising: a microphone arranged proximate a first end of the device; a speaker arranged proximate a second end of the device; a camera arranged between the microphone and the speaker; a radar sensor arranged between the microphone and the speaker; a passive infrared (PIR) sensor assembly arranged between the speaker and the radar sensor, the PIR sensor assembly comprising: a first PIR sensor, a first reflective surface arranged proximate the first PIR sensor, and a second PIR sensor; and a midframe having a first side and a second side, and a printed circuit board assembly (PCBA) disposed on the first side of the midframe and wherein: the first PIR sensor and the second PIR sensor are coupled to the PCBA; and a frame of the PIR sensor assembly is disposed on the second side of the midframe.

5. The device of claim 4, wherein the PIR sensor assembly comprises a frame having: a first cavity, a second cavity, and a wall disposed between the first cavity and the second cavity.

6. The device of claim 5, wherein: the first cavity includes a first top and a first bottom opposite the first top; the second cavity includes a second top and a second bottom opposite the second the first reflective surface couples to at least a portion of the first top and a second reflective surface couples to at least a portion of the second top; the first PIR sensor is oriented in a first direction away from the first bottom; and the second PIR sensor is oriented in a second direction away from the second bottom.

7. The device of claim 6, wherein: the first direction is angled away from the wall; and the second direction is angled away from the wall.

8. The device of claim 5, wherein: the PIR sensor assembly includes a mount having a first pocket and a second pocket; the first PIR sensor includes: a first front at least partially disposed in the first cavity, and a first back disposed within the first pocket; and the second PIR sensor includes: a second front at least partially disposed in the second cavity, and a second back disposed within the second pocket.

9. The device of claim 8, wherein: the mount includes a third front and a third back opposite the third front, the third front being transverse to the third back; the third front includes the first pocket and the second pocket; and the third back is substantially planar.

10. The device of claim 4, wherein the device further includes a second reflective surface arranged proximate the second PIR sensor.

11. The device of claim 10, wherein the first reflective surface and the second reflective surface are surfaces of a first mirror.

12. The device of claim 10, wherein the first reflective surface is a surface of a first mirror and the second reflective surface is a surface of a second mirror.

13. The device of claim 4, further comprising a first housing and a second housing, wherein: the first housing includes first contacts; and the second housing includes second contacts, the second

contacts engaging with the first contacts to at least provide a connection between an electronic component disposed within the first housing and the speaker disposed in the second housing.

14. The device of claim 4, further comprising a printed circuit board (PCB), wherein: the PIR assembly includes: a frame having one or more cavities through which the first PIR sensor and the second PIR sensor receive infrared light, and a mount disposed between the frame and the PCB; and the first PIR sensor and the second PIR sensor are disposed on the mount and are communicatively coupled to the PCB.

15. The device of claim 4, further comprising a housing, wherein the microphone, the speaker, the camera, the radar sensor, and the PIR assembly are disposed within the housing.

16. A device, comprising: a first end; a second end; a printed circuit board assembly (PCBA); a camera coupled to the PCBA; a speaker located closer to the second end than the first end; and a passive infrared (PIR) sensor assembly located farther from the second end than the speaker and farther from the first end than the camera, the PIR sensor assembly comprising: a lens, a frame including a divider, a first PIR sensor coupled to the PCBA, the first PIR sensor being oriented in a first direction away from the divider and towards the first end of the device, a second PIR sensor coupled to the PCBA, the second PIR sensor being oriented in a second direction away from the divider and towards the first end of the device, a first reflective surface coupled to the frame, the first reflective surface arranged to reflect at least some infrared light that passes through the lens towards the first PIR sensor, and a second reflective surface coupled to the frame, the second reflective surface arranged to reflect at least some infrared light that passes through the lens towards the second PIR sensor.

17. The device of claim 16, wherein the first reflective surface and the second reflective surface are surfaces of a first mirror.

18. The device of claim 16, wherein the first reflective surface is a surface of a first mirror and the second reflective surface is a surface of a second mirror.

19. The device of claim 16, wherein: the frame includes: a first cavity, and a second cavity; the first PIR sensor is oriented towards the first cavity; and the second PIR sensor is oriented towards the second cavity.

20. The device of claim 19, wherein: the first reflective surface and the second reflective surface couple to the frame at a location at least partially within the first cavity and at least partially within the second cavity; and the frame includes a wall disposed between the first cavity and the second cavity.

21. The device of claim 16, wherein: the device includes: a first housing, and a second housing configured to couple to the first housing; the first housing includes: the PCBA, the camera, the first reflective surface, the second reflective surface, the first PIR sensor, and the second PIR sensor; and the second housing includes the speaker.

22. The device of claim 16, wherein at least one of: the first PIR sensor is arranged substantially parallel to the first reflective surface; or the second PIR sensor is arranged substantially parallel to the second reflective surface.
